



COURSE MATERIAL

for

CHEMISTRY

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Students are encouraged to complement this study guide with lectures, laboratory practicals, recommended textbooks, and other approved academic resources for a more comprehensive understanding of the course.

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Study Guide

About This Study Guide

This Chemistry Study Guide is designed to assist students in understanding the fundamental concepts and principles of Chemistry. It provides a structured approach to learning the basic ideas about matter, elements, atoms, molecules, compounds, mixtures, chemical naming, and laboratory separation techniques.

The guide contains major topics arranged into study sessions, with each session focusing on an important area of introductory Chemistry. Students are encouraged to study each section carefully, understand the concepts explained, relate them to everyday observations, and practice identifying substances and applying chemical principles.

Course Overview / Course Aim / Course Objectives

This course introduces students to the fundamental concepts of Chemistry and provides a foundation for understanding the composition, structure, properties, and behaviour of matter.

The course aims to help students:

- Understand the meaning, scope, and importance of Chemistry.
- Explain the nature and classification of matter.
- Understand elements, atoms, molecules, and compounds.
- Explain the particulate nature of matter.
- Understand relative atomic masses and their significance.
- Identify and classify chemical compounds.
- Apply basic IUPAC rules in naming chemical compounds.
- Understand mixtures and their properties.
- Develop knowledge of methods used to separate mixtures.
- Relate Chemistry concepts to everyday materials and laboratory activities.

Major Topics Covered

- Introduction to Chemistry.
- Nature and classification of matter.
- Elements and their characteristics.
- Molecules and atomicity.
- Particulate nature of matter.
- Relative atomic masses of elements.
- Compounds and their characteristics.
- IUPAC nomenclature of chemical compounds.
- Mixtures and their properties.
- Practical methods for the separation of mixtures.

Background to the Course

Chemistry is a branch of science concerned with the study of matter, its composition, structure, properties, and the changes it undergoes. Everything around us is made up of matter, including the air we breathe, the water we drink, the food we eat, and the materials used to construct buildings, machines, medicines, and household products.

The study of Chemistry helps students understand what substances are made of and how different substances interact with one another. It also provides the foundation for understanding many areas of science, medicine, agriculture, engineering, environmental science, pharmacy, and technology.

This study material introduces students to the basic ideas required for further study of Chemistry. Students will learn how matter is classified, how elements combine to form compounds, how substances can be identified, and how mixtures can be separated using appropriate physical methods.

Practical activities are also important because they allow students to observe chemical substances and apply theoretical knowledge to real laboratory situations. Through careful observation and experimentation, students can develop scientific reasoning and practical laboratory skills.

Regular study, revision, observation, and practice will help students develop a strong foundation in Chemistry.

Course Study Sessions

Session 1: Introduction to Chemistry

Students will learn the meaning, scope, and importance of Chemistry. The session introduces Chemistry as the study of matter, its composition, properties, structure, and the changes that matter undergoes. Students will also examine the importance of Chemistry in everyday life and its applications in areas such as medicine, agriculture, industry, food production, environmental management, and technology.

Session 2: Nature of Matter

Students will learn about matter and the characteristics that distinguish matter from non-matter. The session introduces the common states of matter and explains how substances can exist as solids, liquids, and gases. Students will examine the basic properties of each state and understand how changes in temperature and pressure can affect the state of a substance.

Session 3: Elements

Students will learn about elements as pure substances that cannot be broken down into simpler substances by ordinary chemical means. The session introduces the names and symbols of elements and explains how elements are represented in Chemistry. Students will also learn about the general classification of elements and their occurrence in the environment.

Session 4: Molecules and Atomicity

Students will learn about molecules and how atoms combine to form molecules. The session

explains the meaning of molecular substances and introduces atomicity as the number of atoms present in a molecule of an element or compound. Students will examine examples of substances with different atomicity and understand how chemical formulae represent the composition of molecules.

Session 5: Particulate Nature of Matter

Students will study the idea that matter is made up of very small particles. The session explains the arrangement and movement of particles in solids, liquids, and gases. Students will examine how particle movement accounts for properties such as diffusion, expansion, changes of state, and the mixing of substances.

Session 6: The Relative Atomic Masses of Elements

Students will learn about the relative masses of atoms and the concept of relative atomic mass. The session explains how the masses of atoms are compared using a standard reference and introduces the use of relative atomic masses in chemical calculations. Students will also learn how relative atomic mass values are obtained and how they are used when working with chemical formulae.

Session 7: Compounds

Students will learn about compounds as substances formed when two or more different elements combine chemically in definite proportions. The session explains the differences between elements and compounds and examines the properties of compounds. Students will also learn how chemical formulae are used to represent compounds and how compounds can be formed through chemical reactions.

Session 8: IUPAC Nomenclature of Chemical Compounds

Students will learn the basic principles used to name chemical compounds according to the International Union of Pure and Applied Chemistry (IUPAC) system. The session introduces the naming of common ionic and covalent compounds and helps students understand how chemical names are connected to chemical formulae. Students will practise identifying compounds from their formulae and writing appropriate chemical names.

Session 9: Mixtures

Students will learn about mixtures and how they differ from compounds. The session explains that mixtures contain two or more substances that are physically combined and can usually be separated by physical methods. Students will examine different types of mixtures and consider examples found in everyday life, including mixtures of solids, liquids, and gases.

Session 10: Practical on Separation of Mixtures

Students will learn about practical methods used to separate the components of mixtures. The session introduces techniques such as filtration, evaporation, crystallization, distillation, sublimation, decantation, sedimentation, magnetic separation, and chromatography. Students will learn how the physical properties of substances determine the appropriate separation method and will apply these principles during laboratory activities.

Recommended Study Approach

Students should:

- Read each topic carefully.
- Understand important Chemistry concepts and explanations.
- Learn the meanings of important chemical terms.
- Study the symbols and formulae of common substances.
- Practice naming chemical compounds.
- Relate Chemistry concepts to everyday materials.
- Review the properties of elements, compounds, and mixtures.
- Study the principles behind different separation methods.
- Carry out practical activities carefully and safely.
- Revise each study session regularly.
- Attempt examination-style questions after completing each topic.

Consistent study, revision, observation, and practical work will help students develop a strong understanding of introductory Chemistry and prepare effectively for further Chemistry studies and examinations.

STUDY SESSION ONE: Chemistry as a Subject and as a Central Science

1.0 INTRODUCTION

This session provides readers with a detailed explanation of the fundamental concepts of Chemistry, familiarization with the chemistry laboratory, matter, its properties, and classification. It introduces the role of chemistry as the central science bridging physical, life, and applied sciences.

The session also explains the scope of chemistry, its core branches—including organic, inorganic, physical, analytical, and biochemistry—and how chemical principles govern everyday phenomena, biological systems, and industrial applications. In addition, it highlights the importance of the scientific method, chemical composition, and systematic investigation. Furthermore, the session introduces basic chemistry laboratory equipment, their uses, measurement techniques, and safety practices required for effective laboratory investigations.

1.1 LEARNING OUTCOMES FOR THE STUDY SESSION 1

When you have studied this session, you should be able to:

- i. Define chemistry and explain its scope as a scientific discipline.
- ii. Identify and describe the major branches of chemistry.
- iii. Explain why chemistry is considered the central science connecting various scientific fields.
- iv. Distinguish between fundamental chemical classifications of matter, including elements, compounds, and mixtures.
- v. Relate chemical principles to biological, environmental, and industrial applications.
- vi. Identify common chemistry laboratory equipment and their specific uses.
- vii. Explain basic laboratory safety procedures and hazard handling practices.

1.2 CHEMISTRY AS A DISCIPLINE:

Driven by an innate curiosity about natural phenomena, human beings have always sought to investigate the world through careful observation and experimentation. This ongoing spirit of inquiry has fueled global research efforts and generated a vast body of empirical data.

When this information is organized systematically for the benefit of humankind, it forms what we know as **science**. Due to its massive expansion and diverse fields, science is categorized into several distinct branches—such as Physics, Biology, Geology, and Chemistry. Among these, chemistry stands out as one of the most vital and foundational scientific disciplines, and it serves as the primary focus of this text. Chemistry may be defined as the branch of science which deals with the study of matter, its composition, its properties and the changes which it undergoes in composition as well as in energy during various processes.

The word chemistry has been derived from the word alchemy, which means ‘study of met also Alchemy itself might have come from al chemical marked effect on our present day life.

Chemistry has helped us to meet all our requirement for better living. The continuous research in the field of chemistry has resulted in the production of useful materials such as, clothes, drugs, artificial foods, plastics, rubbers, fertilizers, insecticides, weed killers, lifesupporting products, etc., which have revolutionized our life.

Our life would have been very dull and dreary without the knowledge of chemistry. In fact, we can say, chemistry is everywhere in the world around us; it is, in what we eat; in what we breathe; in how we live and even in what we are.

1.3 CHEMISTRY A CENTRAL SCIENCE

Modern chemistry is both an intellectually challenging conceptual science and a deeply practical field that lies at the core of humanity's future. As a central discipline, chemistry connects nearly every branch of science, providing vital tools to investigate biological, physical, medical, and environmental phenomena. Chemists collaborate closely with professionals across diverse sectors:

- Biologists to decode life processes and metabolic functions.
- Physicists to analyze the properties of matter and engineer innovative energy sources.
- Geologists to investigate both terrestrial and cosmic environments.
- Physicians to design advanced drugs and therapeutic medications.
- Ecologists to monitor and improve environmental health.
- Engineers and technical managers to supply the materials and energy necessary for a higher quality of life.

Ultimately, chemistry responds to broad societal needs, playing a crucial role in discovering new industrial processes, harvesting alternative energy sources, developing advanced materials, supporting global food security, advancing healthcare, and protecting our environment.

1.4 BRANCHES OF CHEMISTRY

Chemistry can be broadly divided into Pure Chemistry and Applied Chemistry.

A. PURE CHEMISTRY

The vast domain of chemical science is generally categorized into two complementary realms: **Pure Chemistry** and **Applied Chemistry**. While pure chemistry seeks to expand our fundamental understanding of nature and molecular behavior, applied chemistry translates those theoretical insights into practical innovations that directly benefit society.

A. Pure Chemistry: Unlocking Theoretical Foundations

Pure chemistry is driven by curiosity and the pursuit of fundamental truths about how matter is structured and transforms. It is traditionally anchored in three primary pillars, each branching into a wide array of specialized sub-disciplines:

- **Organic Chemistry:** This branch investigates the structural properties, synthesis, and reactivity of carbon-based compounds—specifically hydrocarbons and their various derivatives. Because carbon forms the backbone of all living systems, polymers, and modern pharmaceuticals, organic chemistry explains the complex molecular architecture essential to life.

- **Inorganic Chemistry:** Focusing on compounds outside the organic classification, this branch examines minerals, metals, organometallic complexes, and non-living matter predominantly found in the Earth's crust. It provides crucial insights into crystal structures, coordination chemistry, and solid-state materials.
- **Physical Chemistry:** This discipline establishes the theoretical framework governing chemical systems. By integrating principles of physics, thermodynamics, quantum mechanics, and chemical kinetics, physical chemistry explains *why* and *how* chemical transformations occur at energetic and molecular levels.

B. Applied Chemistry: Driving Practical Innovation

Applied chemistry leverages chemical principles and experimental techniques to solve real-world challenges, manufacture useful products, and improve human life. Its major specialized branches include:

- **Analytical Chemistry:** Utilizing a sophisticated toolkit of separation, identification, and measurement techniques, analytical chemistry determines the precise composition of substances. It is divided into two vital practices:
- **Industrial Chemistry:** This field focuses on the large-scale, commercial manufacturing of chemical products. It bridges laboratory discoveries with industrial-scale engineering to produce everyday commodities, fertilizers, acids, and polymers safely and efficiently.
- **Biochemistry:** Exploring the chemical basis of life, biochemistry examines the molecular, cellular, and metabolic pathways operating within living organisms. It details how enzymes catalyze reactions, how genetic codes are expressed, and how cellular energy is managed.
- **Geochemistry:** This branch investigates the chemical composition and dynamics of Earth. It analyzes soil chemistry, mineral formation, rock metamorphism, and the natural geological conditions responsible for fossil fuel genesis.
- **Petrochemistry:** A specialized division of industrial chemistry dedicated to refining and transforming crude oil and natural gas into valuable fuels, lubricants, solvents, and foundational raw materials for the plastics and chemical manufacturing industries.
- **Radiochemistry.** It is a branch of chemistry which deals with the study of radioactive materials, both natural as well as man-made. It also involves the use of radioactive materials to study the pathways/mechanism of ordinary chemical reactions.
- **Biotechnology.** It refers to the technological applications which uses biological systems, living organisms. Biotechnology, in fact, combines various disciplines like genetics, molecular biology, biochemistry, embryology and cell biology for developing techniques for beneficial effects.
- **Medicinal or Pharmaceutical Chemistry.** It is scientific discipline at the intersection of chemistry and pharmacology, which is involved with designing synthesizing and developing pharmaceutical drugs. Medicinal chemistry involves the identification, synthesis and

development of new chemical entities suitable for therapeutic use. Medicinal chemistry is highly interdisciplinary science that combines organic chemistry with biochemistry, pharmacology, pharmacognosy; molecular biology, statistics and physical chemistry.

- **Environmental Chemistry.** It is a branch of chemistry which deals with scientific study of chemical and biochemical phenomena that occur in natural places. Environmental chemistry is also an interdisciplinary science that includes atmospheric, aquatic, and soil chemistry along with analytical chemistry, environmental studies and other areas of science.

1.5 CAREER OPPORTUNITIES

Since chemistry is a central science because of its multidisciplinary nature, therefore, chemistry students can pursue their careers in the field of industries, education, research work, government agencies and other non-radiation fields. Some of the careers opportunities, that a student with degree in chemistry can have are as follows:

1. **Industries.** Chemical industries employ about 66% of all the chemists. The majority of them find opportunity in research and product development (Rand D), sales, or marketing. Many of them work in quality control analysis and testing products. Other find work in areas like industrial hygiene and safety or regulatory work for environmental compliance. Pharmacognosy is a study of medicines derived from natural
2. **Academic Institutions.** Educational institutions employ about 26% of the chemists. Ph. D. degrees are required for most of academic positions at the colleges and universities. Some of chemists having graduation degree in education take up teaching assignments in high schools.
3. **Government Agencies.** Government-employ about 7Cfr: of all the chemists. Federal, local and state Government agencies hire chemists for variety of jobs including basic research, testing work required to enforce government regulations, technical program managers, authors/ editors of technical documents and government regulations.
4. **Non-traditional Fields.** A small percentage of chemists (about 1%) find work in non -traditional fields. They get opportunities to become patent lawyers, science writers, information specialists, technical librarians, technical consultants or business owners.

1.6 Group Discussion

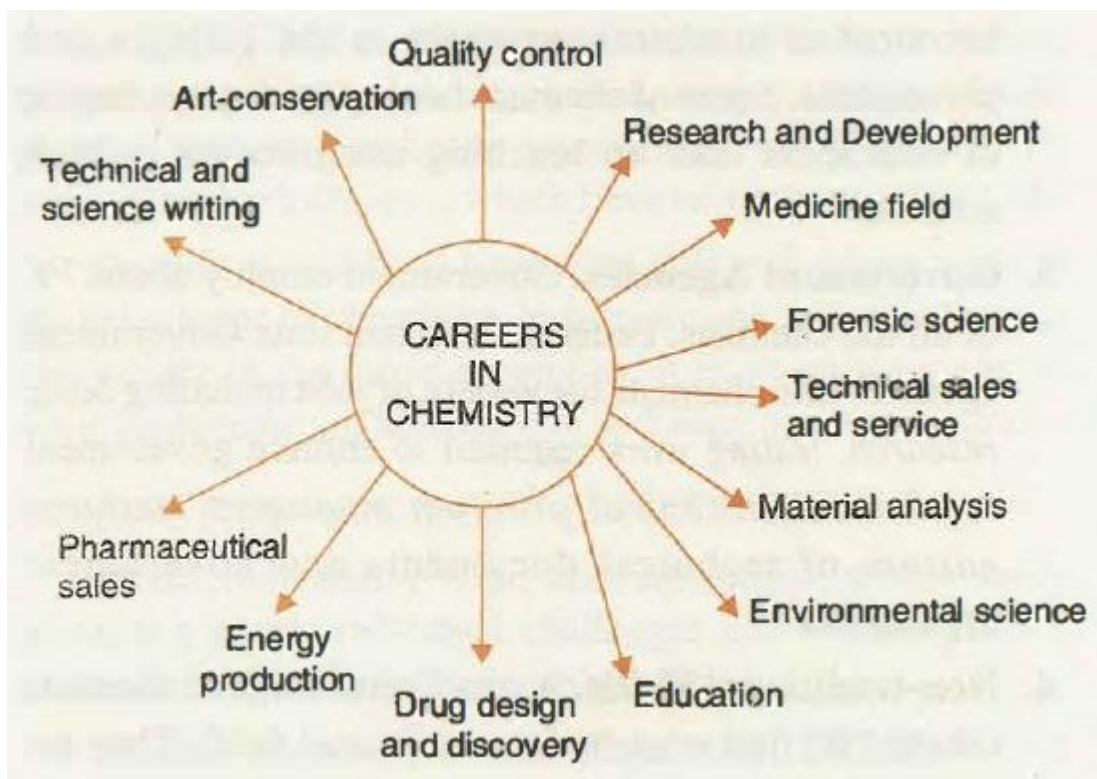
Identify two applied chemistry professions and explain the chemistry they practice.

Hints: For reference, the two important professions associated with applied chemistry are being discussed as follows:

1. **Analyst:** Analysts find jobs in chemical industry, food industry and pathological laboratories. In chemical industries, they control the quality of final product through chemical analysis of the product. In food laboratories, they analyze the food items to detect adulteration. In pathological labs they carry out chemical tests on sample of blood or urine to help the doctor for diagnosis of disease.
2. **Research scientist.** Research scientists find jobs in pharmaceutical companies where they can use their knowledge of chemistry in developing more convenient and

economical methods for the synthesis of drugs. They can also help in designing new drugs.

Different career options in chemistry are summarized in Fig. 11



1.7 Various careers associated with chemistry.

At its core, **chemistry** is the comprehensive study of matter—encompassing its atomic composition, physical and chemical properties, and the structural transformations it undergoes. Crucially, chemistry also examines the energy changes that accompany these reactions, determining how thermal, electrical, or radiant energy is absorbed or released during chemical processes.

As the **central science**, chemistry serves as the foundational bridge connecting diverse scientific disciplines like biology, physics, earth science, and medicine. To manage and explore this vast field effectively, chemistry is broadly divided into two major domains: **Pure Chemistry** and **Applied Chemistry**.

A. Pure Chemistry: Uncovering Fundamental Principles

Pure chemistry focuses on expanding our theoretical understanding of natural laws and molecular behavior. It is anchored by three primary branches:

- **Organic Chemistry:** The study of carbon-containing compounds, hydrocarbons, and their diverse functional derivatives.
- **Inorganic Chemistry:** The investigation of non-carbon compounds, minerals, and metals that comprise the non-living elements of the Earth.
- **Physical Chemistry:** The exploration of the energetic laws, thermodynamics, and quantum mechanical principles that govern chemical systems and transformations.

B. Applied Chemistry: Translating Science into Solutions

Applied chemistry harnesses theoretical insights to solve practical problems, drive technological innovation, and improve human welfare. Its diverse branches include:

- **Analytical Chemistry:** The practice of qualitative and quantitative measurement to determine the exact chemical makeup and concentration of various materials.
- **Biochemistry:** The study of the molecular, cellular, and metabolic pathways that sustain living organisms.
- **Geochemistry:** The examination of chemical reactions, mineral compositions, and elemental cycles occurring within the Earth's crust and interior.
- **Petrochemistry:** The industrial transformation of crude oil and natural gas into high-value fuels, plastics, and essential chemical raw materials.
- **Environmental Chemistry:** The study of chemical phenomena in nature, focusing on pollutant pathways, atmospheric reactions, water quality, and the development of sustainable "green chemistry" practices.
- **Radiochemistry:** The investigation of radioactive elements, nuclear isotopes, and their chemical behaviors, with vital applications in medical diagnostics, cancer treatment, and nuclear energy generation.
- **Biotechnology:** The interdisciplinary integration of chemistry, biology, and engineering to manipulate biological systems and cellular components for developing advanced pharmaceuticals, agricultural solutions, and renewable biofuels.

EVALUATION

1. Which branch of pure chemistry deals with the study of compounds associated with non-living sources?

- (a) Physical chemistry (b) Organic chemistry
- (c) Biochemistry (d) Inorganic chemistry

2. Which of the following is not a applied chemistry?

- (a) Geochemistry (b) Biochemistry

(c) Radiochemistry (d) inorganic chemistry.

3. In order to design new drug, a chemist has to seek the help of

(a) Engineer (b) Geologist

(c) Zoologist (d) Physician.

4. The branch of chemistry which deals with the study of hydrocarbons is called

(a) Organic chemistry (c) Radiochemistry

(b) Inorganic chemistry (d) Nuclear chemistry.

II. Fill in the Blanks

5. Complete the following sentences by supplying appropriate words:

(i) Ecologist and chemist work together to

(ii) The branch of pure chemistry which deals with study of fundamental laws and principles is called

(iii) Qualitative analysis deals with _____

(iv) Radiochemistry deals with the study of _____ substances.

(v) The phenomenon of metamorphosis of rocks is studied by _____ branch of chemistry.

1.8 THE SCIENTIFIC METHOD

1.8 The Scientific Method: The Engine of Empirical Discovery

In the pursuit of knowledge, scientists rely on a systematic, objective framework known as the **scientific method**. Far from a rigid linear formula, it is a dynamic, self-correcting process designed to gather measurable, empirical evidence, eliminate human bias, and test explanations of natural phenomena.

Whether exploring chemical reactions, physical forces, or biological systems, the scientific method provides a rigorous pathway to transform raw observations into reliable scientific theories.

Key Steps of the Scientific Method

While investigations can take many forms depending on the discipline, the scientific method generally unfolds through a structured sequence of logical steps:

- **Make Observations:**

The journey begins with careful, objective perception of the natural world. Observations can be **qualitative** (descriptive, such as color changes or texture) or **quantitative** (numerical

measurements like temperature, mass, or volume) obtained using senses or precision instruments.

- **Ask Questions and Gather Information:**

Anomalies or patterns in observations prompt specific questions (e.g., “*Why does this reaction speed up when heated?*”). Scientists then review existing literature and historical data to understand what is already known about the phenomenon.

- **Formulate a Hypothesis and Predict:**

A hypothesis is a tentative, testable, and falsifiable explanation for the observed phenomenon. It is frequently framed as an “**If... then... because...**” statement that links a proposed cause to a measurable effect, generating clear predictions.

- **Design and Execute Controlled Experiments:**

To test the hypothesis, scientists conduct meticulously designed experiments. These must isolate variables—controlling constants while manipulating an **independent variable** to measure its effect on a **dependent variable**—ensuring the procedure can be replicated independently by other researchers.

- **Analyze Data and Draw Conclusions:**

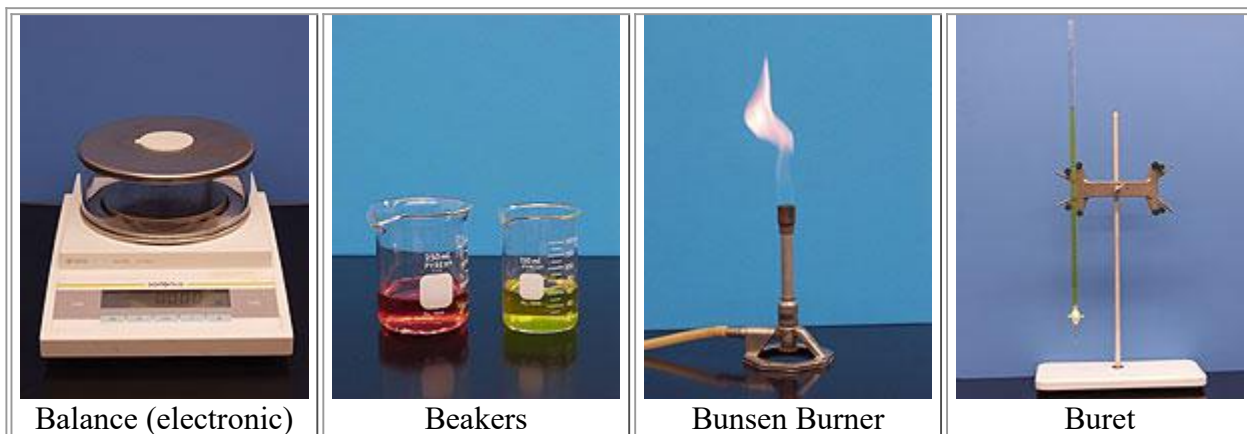
Experimental results yield raw data, which is organized, visualized, and evaluated using statistical methods. Based on this evidence, the scientist determines whether to **accept, reject, or modify** the original hypothesis. If the data contradicts the hypothesis, it is refined for further testing.

- **Replication, Peer Review, and Theory Building:**

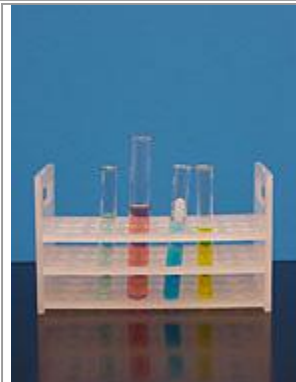
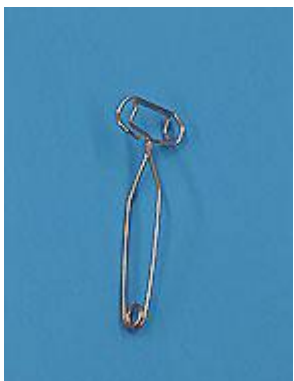
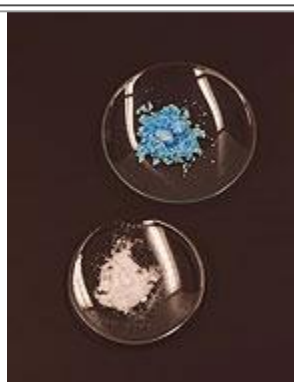
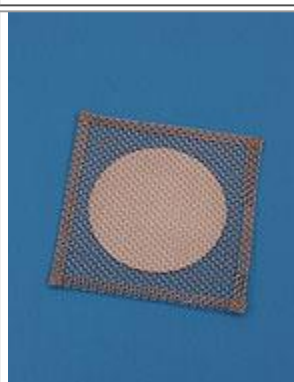
Science thrives on reproducibility. An experiment must be replicated across multiple independent laboratories until consistent results eliminate discrepancies between observations and theory. Over time, rigorously validated hypotheses converge to form broader, robust **scientific laws and theories**

1.9 CHEMISTRY LABORATORY COMMON EQUIPMENT

Below are photos and names of common lab equipment you will encounter in Chemistry.



			
Clay Triangle	Crucible	Crucible in Triangle	Crucible Tongs
			
Dropper Pipets	Dropper in action	Erlenmeyer Flasks	Evaporating Dish
			
Forceps	Funnels	Goggles	Graduated Cylinders
			
Pinch Clamp	Pipets and Bulbs	Plastic and Rubber Policemen	Ring Clamp & Stand

			
Scoopula	Stirring Rods	Thermometers	Test Tubes in Rack
			
Test Tube Holder	Tube & Holder in Action	Utility Clamp	Clamp in action
			
Wash Bottle	Watch Glasses	Wire Gauze	Combined for Heating

A List of Chemistry Laboratory Apparatus and Their Uses

FUNCTIONS OF COMMON LABORATORY EQUIPMENT

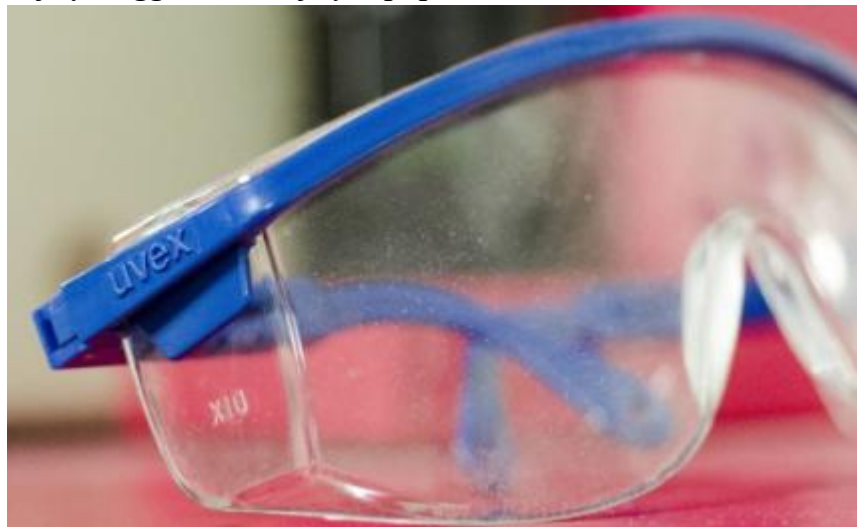


Functions of common pieces of laboratory equipment.

In most labs, you'll encounter the same basic apparatus. Here, the use for each is explained. You will learn about:

- Safety goggles and safety equipment
- Beakers
- Erlenmeyer flasks, AKA conical flasks
- Florence flasks, AKA boiling flasks
- Test tubes
- Watch glasses
- Crucibles
- Funnels
- Graduated cylinders
- Volumetric flasks
- Droppers
- Pipettes
- Burets
- Ring stands, rings, and clamps
- Tongs and forceps
- Spatulas and scoopulas
- Thermometers
- Bunsen Burners
- Balances

Safety Goggles and Safety Equipment



The first and foremost rule of any laboratory is to be safe! This may seem obvious, but people often disregard safety protocols for one reason or another, putting themselves and those around them in danger. The best thing you can do is to make sure you follow all safety protocols at all times. Safety goggles are required wear in all chemistry labs. Not wearing them puts you in danger of eye irritation and possibly blindness in the case of an accident. A small droplet of acid could splash out of the container at any time. Better safe than permanently blinded!

Latex gloves should be used when there is a possibility of corrosive chemicals spilling onto your hands.

A lab apron or coat can also prevent injury in case of spills or splashes.

Never wear open-toed shoes or sandals in a lab.

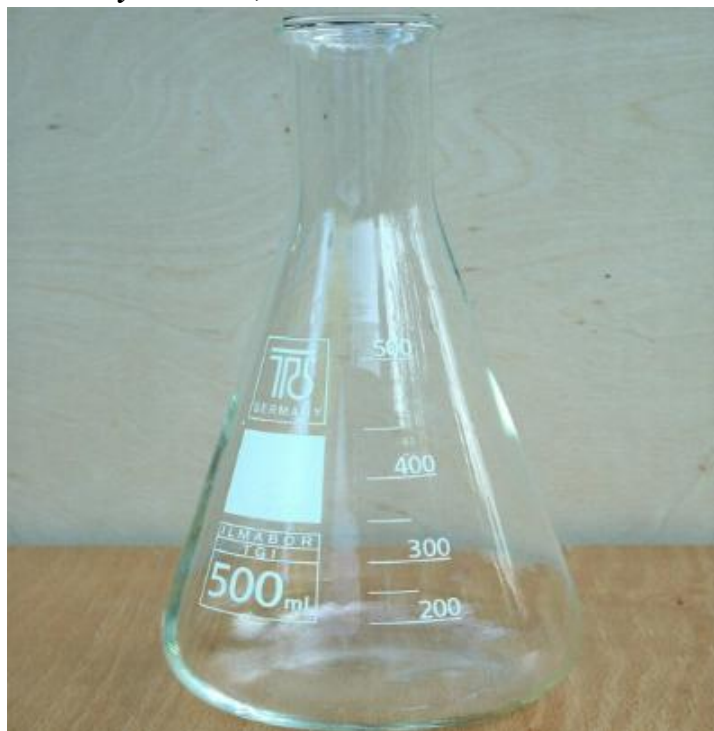
Beakers



A beaker is a common container in most labs. It is used for mixing, stirring, and heating chemicals. Most beakers have spouts on their rims to aid in pouring. They also commonly have lips around their rims and markings to measure the volume they contain, although they are not a precise way to measure liquids. Beakers come in a wide range of sizes.

Because of the lip that runs around the rim, a lid for a beaker does not exist. However, a watch glass can be used to cover the opening to prevent contamination or spillage.

Erlenmeyer Flasks, AKA Conical Flasks



Erlenmeyer flask.

Also known as a conical flask, the Erlenmeyer flask was named after its inventor in 1861. It has a narrow neck and expands toward its base. This allows easy mixing and swirling of the flask without too much risk of spilling. The narrow opening also allows for the use of a rubber or glass stopper. It can easily be clamped to a ring stand as well as heated or shaken mechanically.

Once again the marks on the side are meant primarily for estimation rather than precision.

An important safety tip here is to never heat this flask while it is capped. This could cause a pressure build-up that could result in explosion.

Florence Flasks, AKA Boiling Flasks

Also known as a boiling flask, the Florence flask has a round bottom and a long neck. It is used to hold liquids and can be easily swirled and heated. It can also easily be capped by rubber or glass stoppers.

Once again, safety dictates that this flask never be heated when capped. Pressure build-up and explosions can and do occur.

Test Tubes



Test tubes held in spring clamps.

A test tube is a glass tube with one end open and the other end closed. The closed end is rounded. Test tubes are used to hold small samples. They are primarily used for qualitative assessment and comparison. A common place to see these is the biochemistry lab. When a large number of samples need to be tested and compared, test tubes are used to make this easier. They are also easily capped with a rubber or glass stopper.

They are generally held in a test tube rack specifically designed for the purpose. If heated or unsafe to touch with bare hands, test-tube tongs can be used to move them.

Never heat a capped test tube.

Watch Glasses



A watch glass holding a powder.

A watch glass is just a round piece of glass that is slightly concave/convex (think of a lens). It can hold a small amount of liquid or solid. They can be used for evaporation purposes and also can function as a lid for a beaker.

CRUCIBLES:

A crucible is a small clay cup made of a material that can be heated to extreme temperatures. This is because they are used for heating. They come with lids.

Funnels



An inverted funnel positioned above a watch glass.

A lab funnel is just like any other funnel except that it was designed to be used in a laboratory setting. They can be made of plastic or glass and can have either a short stem or a long stem, depending on what they are needed for. There are several sizes that can be chosen from based on the amount of liquid that needs to go through them quickly.

Graduated Cylinders



Graduated cylinders.

This is a primary measuring tool for the volume of a liquid. There are several markings up and down the length of the container with specific increments. Graduated cylinders come in many sizes. The smaller they are, the more specific the volume measurements will be.

When reading the volume from a graduated cylinder, you will notice that the liquid seems to have an indentation. The liquid around the edges will be higher than the liquid in the center, sloping down like the sides of a trampoline when someone is standing in the middle. This is called the meniscus. Line the lowest point of the meniscus up with the nearest marking, keeping the cylinder level. That is how to properly read the volume

Volumetric Flasks

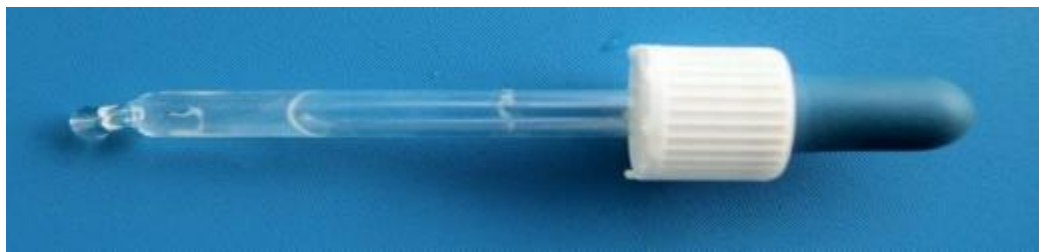


A 500-ml volumetric flask.

A volumetric flask is a round flask with a long neck and flat bottom. It is used to measure an exact volume of liquid. There is a small line on the neck that indicates how far to fill the bottle (Use the bottom of the meniscus). They come with special caps that will not let anything in or out.

Remember that temperature affects volume; therefore avoid using liquids that will fluctuate in temperature (hot water that will cool, for example).

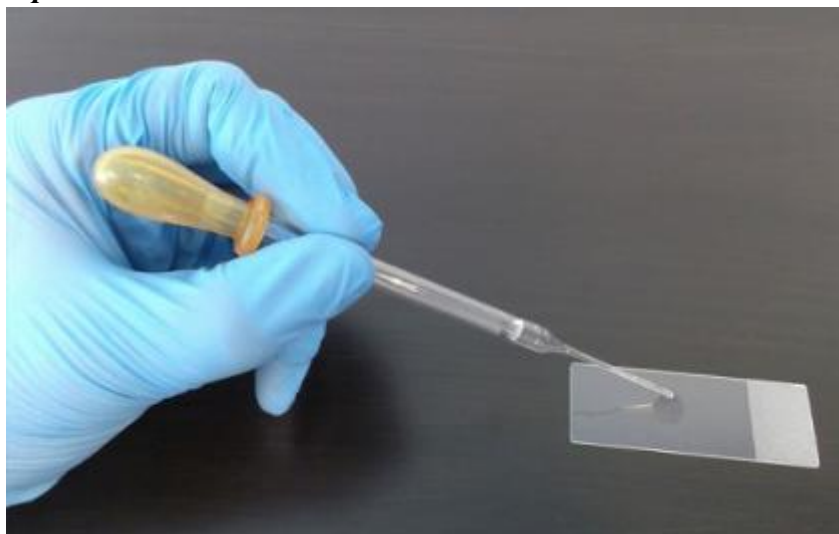
Droppers



A glass dropper.

These are small glass tubes with narrow tips and a rubber bulb on the end. They suck up liquid that can then be squeezed out in small drops. These can be used to add an indicator to a solution about to be titrated.

Pipettes



A Pasteur pipette.

There are a large variety of pipettes designed to accomplish specific goals. However, they are all for measuring an exact volume of liquid and placing it into another container.

Burette



A Mohr burette.

A burette is a glass tube that is open at the top and comes to a narrow pointed opening at the bottom. Right above the bottom opening is a stopcock that can be turned to control the amount of liquid being released. There are markings along the length of the tube that indicate the volume of liquid present.

A burette is used for extremely accurate addition of liquid. By adjusting the stopcock, the amount of liquid that is released can be slowed to a drop every few seconds. Burets are one of the most accurate tools in the lab. Burettes are set up by using a burette clamp in combination with a ring stand, discussed below.

To determine how much liquid is added, write down how much is initially in the burette. Then when you're finished adding, write down how much is left. Subtract the final amount from the initial amount and you have the volume of liquid added.

Remember to measure from the bottom of the meniscus!

Ring Stands, Rings, and Clamps

The ring stand is used to suspend burets, beakers, flasks, crucibles, etc. above other containers or in some cases a heat source (Bunsen burner, discussed below).

Always make sure everything is clamped to the stand tightly. When clamping glass, be careful not to shatter the glass. Only tighten that end until snug.

When using a ring on the stand, there are usually other pieces necessary to accomplish the goal. Wire mesh is laid across the ring to distribute evenly heat and support the beaker. A clay triangle with an open center is used to suspend crucibles.

Make sure everything is balanced! Do not let the whole setup tip over.

Tongs and Forceps



Forceps.

Tongs and forceps are for grabbing things that should not be touched by hand. Some tongs are specially made to hold beakers, others to hold test tubes, and so on. There are also general tongs.

Forceps are used to grab small things like solid chemicals that are broken into chunks, so they can be safely handled and added to containers.

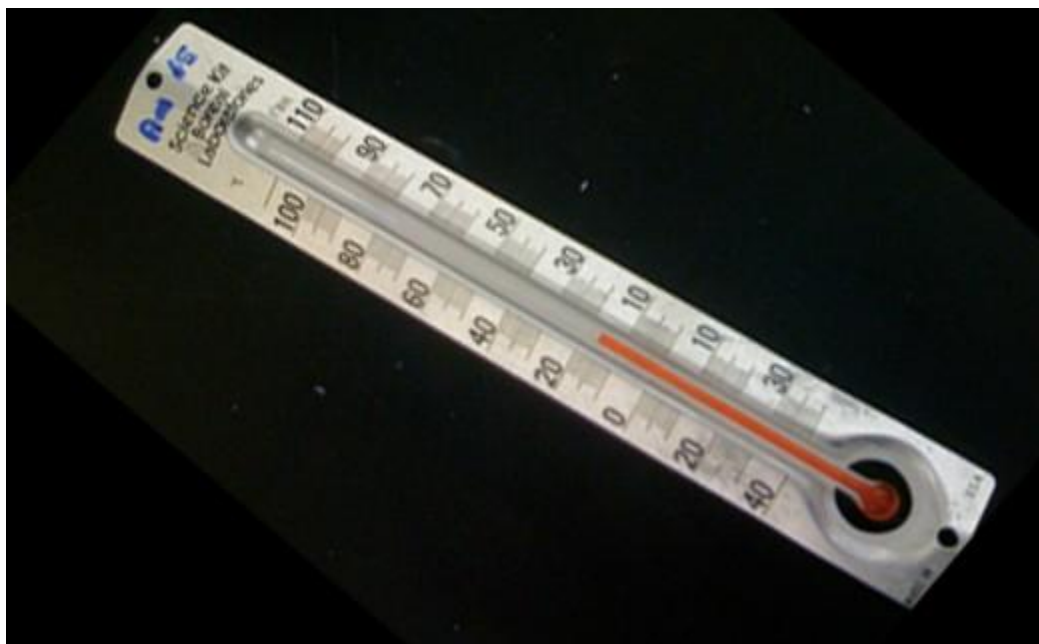
Spatulas and Scoopulas



Two scoopulas.

Spatulas and scoopulas are for scooping solid chemicals. The typical use for these in a lab is scooping chemical out of its original container onto a weigh boat so that it can be weighed on a balance.

Thermometers



Glass thermometer

A laboratory thermometer is a glass thermometer used for measuring the temperature of liquids.

Bunsen Burners



A lit Bunsen burner.

A Bunsen burner is a mechanical apparatus that is connected to a flammable gas source. There is a knob to adjust the amount of gas flow and a rotating collar that controls airflow. These both must be adjusted to get an ideal flame for heating purposes. The burner is lit with a striker. Utmost safety is required when using a Bunsen burner.

Balances



Triple beam balance.

A balance is used to weigh chemicals. The chemicals are always in some form of container and never placed directly on the balance. It is important not to move a balance because they have been calibrated for the exact position they are in. Some balances have plastic housing with small doors to keep air currents from affecting the measurement. Close these doors whenever the balance is in use. To use a balance to determine the weight of a chemical, first put the empty container that the chemical will be in on the balance. Once you have a reading, press the "tare" or "zero" button on the balance. Remove the container from the balance and add the chemical (never add chemicals to a container while it is on the balance). Reweigh after adding the chemical to find the weight of only the chemical. It is important to keep the balance clean.

EVALUATION

1. Which branch of pure chemistry deals with the study of compounds associated with non-living sources?

- (a) Physical chemistry (b) Organic chemistry
(c) Biochemistry (d) Inorganic chemistry

2. Which of the following is not an applied chemistry?

- (a) Geochemistry (b) Biochemistry
(c) Radiochemistry (d) inorganic chemistry.

3. In order to design a new drug, a chemist has to seek the help of

- (a) Engineer (b) Geologist
(c) Zoologist (d) Physician.

4. The branch of chemistry which deals with the study of hydrocarbons is called

- (a) Organic chemistry (c) Radiochemistry
(b) Inorganic chemistry (d) Nuclear chemistry.

II. Fill in the Blanks

5. Complete the following sentences by supplying appropriate words:

- (i) Ecologist and chemist work together to
(ii) The branch of pure chemistry which deals with study of fundamental laws and principles is called
(iii) Qualitative analysis deals with
(iv) Radiochemistry deals with the study of substances.
(v) The phenomenon of metamorphosis of rocks is studied by branch of chemistry.

STUDY SESSION TWO: PARTICULATE NATURE OF MATTER

2.0 INTRODUCTION

Understanding Matter: Its States and Particulate Nature

Every day, we interact with countless objects that we can see, touch, smell, taste, or hear. From the water we drink to the air we breathe, everything around us has mass, takes up space, and can be perceived by our senses. In science, we call all of this **matter**.

1. How Do We Classify Matter?

Matter is incredibly diverse, but we can organize it in two main ways: by its **physical state** and by its **chemical nature**.

- **By Physical State:**
 - **Solids:** Rigid structures with fixed shapes and volumes (e.g., sand, rocks, iron).
 - **Liquids:** Fluids that take the shape of their container while maintaining a fixed volume (e.g., water, oil).
 - **Gases:** Highly fluid substances that expand to fill any container completely (e.g., air, oxygen).
- **By Chemical Nature:**
 - **Elements and Compounds:** Pure substances made of a single type of atom or chemically bound molecules.
 - **Mixtures:** Combinations of two or more pure substances that are physically blended together.

2.1 The Particulate Nature of Matter

Matter may appear smooth and continuous to the naked eye, but scientific evidence shows that it is actually made up of tiny, separate particles. Experiments have demonstrated that these particles have four important characteristics:

1. Particles Are Extremely Small

Consider a small crystal of potassium permanganate dissolved in water. If a few drops of this solution are repeatedly diluted in several beakers of water, the water may still retain a faint purple colour. This demonstrates that even a tiny crystal contains millions of extremely small particles that can spread throughout the water.

2. There Are Spaces Between Particles

When sugar is dissolved in water, the sugar crystals seem to disappear, yet the water level does not increase significantly. This happens because the tiny sugar particles fit into the microscopic spaces between the water particles.

3. Particles Are Constantly Moving

Have you ever noticed how the smell of perfume or freshly cooked food spreads throughout a room? This occurs because particles are constantly moving and spreading from one place to another.

- When a drop of ink is added to water, it naturally spreads out until the liquid is uniformly colored.

- **Temperature Matters:** Heating speeds up this process. If you drop potassium permanganate into hot water versus cold water, it spreads much faster in the hot water because higher temperatures give particles more kinetic energy, making them move more rapidly.
- This spontaneous intermixing of different substances through particle movement is known as **diffusion**.

IV. Particles Attract Each Other

Particles do not just float apart; they hold together because of forces of attraction. This cohesive force is strong enough in gases that, under high pressure, they can be compressed and liquefied.

3. What Are the Building Blocks of Matter?

Depending on the substance, the fundamental constituent particles of matter generally fall into three categories:

Particle Type	Description & Examples
Atoms	Individual, unbounded building blocks (e.g., Argon, Neon, Helium, Diamond).
Molecules	Groups of atoms chemically bonded together (e.g., Sucrose, Glucose, Urea, Methane, Carbon dioxide).
Ions	Charged particles formed when atoms gain or lose electrons (e.g., Sodium chloride, Magnesium oxide, Zinc sulphide).

2.2 STATES OF MATTER

Matter can be classified into three categories depending upon its physical state, namely: solids, liquids and gases. These states of matter arise due to variation in the characteristics of the particles of matter.

2.2.1 PROPERTIES OF SOLIDS

- The matter in solid state possesses a definite volume, a definite shape, distinct boundaries and a definite mass.
- Solids are rigid and almost incompressible.
- Solids may break under force but it is difficult to change their shape.

(iv) Solids generally possess high densities.

(v) Solids do not exhibit diffusion. Some common examples are: table, chair, common salt, silver, ice, diamond, etc

2.2.2 PROPERTIES OF LIQUIDS

(i) The matter in liquid state possesses a definite volume, a definite mass, but no definite shape.

(ii) Liquids are also almost incompressible but are not rigid. In fact, they can flow and acquire the shape of the container in which they are kept.

(iii) Liquids can undergo diffusion.

(iv) Liquids also have high densities but less than that of solids.

Some examples are: milk, water, alcohol, petrol, kerosene, fruit juices, etc.

2.2.3 PROPERTIES OF GASES

i. The matter in gaseous state has neither definite volume nor definite shape but it has definite mass. It acquires the shape and volume of the container.

ii. Gases are highly compressible. For example, natural gas in compressed form is used as fuel (Compressed Natural Gas-CNG) in internal combustion engines. Oxygen supplied to hospitals in cylinders is also in compressed form. Due to high compressibility large volumes of gas can be compressed into a small cylinder and transported easily.

iii. The gases exhibit the property of diffusing very fast into other gases.

iv. Gases exert pressure on the walls of the container in which they are stored.

v. Gases have very low densities.

Some common examples of gases are: air, hydrogen carbon(IV) oxide, hydrogen, Sulphide, ammonia, oxygen, nitrogen, etc.

- **In solids**, the spaces between particles are very small, and the particles have relatively low energy compared with those in liquids and gases. As a result, the particles cannot overcome the strong forces of attraction holding them together. Instead, they can only vibrate around their fixed positions. Therefore, solid particles have vibrational and rotational motion but no translational motion. The small spaces between particles make solids almost incompressible, while the lack of translational motion gives them rigidity.

- **In liquids**, the spaces between particles are larger than those in solids, and the particles possess more energy. This allows them to partially overcome the forces of attraction between neighbouring particles and move more freely. However, the attractive forces are still strong enough to keep the particles close together within the liquid. Liquid particles therefore exhibit vibrational, rotational, and translational motion.

- **In gases**, the spaces between particles are very large, and the particles have much more energy than those in solids and liquids. They have enough energy to overcome the attractive forces between them almost completely. Consequently, gas particles move rapidly and randomly in all directions, spreading to occupy all available space. This is why gases have neither a fixed shape nor a fixed volume and completely fill the container in which they are placed. As gas particles move freely, they constantly collide with one another and with the walls of the container. The pressure exerted by a gas is produced by these collisions with the container walls.

- Solid and liquid states are known as condensed states of matter due to smaller inter-particle spaces” and negligible compressibility.
- Liquids and gases are known as fluids because of their ability to flow and take the shape of container

2.2.4 PLASMA STATE-The Fourth State of matter

The matter in this state is in the form of ionized gas. It consists of neutral mixture of positive ions and unbound electrons. The matter exists in this state at temperatures in the range 10000°C to 15000°C. The matter in the sun and stars exists in plasma state. It is estimated that 99% of the matter in the universe exists in plasma state. Neon in neon lights is also in plasma state.

Plasma

- Plasma has neither a definite volume nor a definite shape.
- Plasma often is seen in ionized gases. Plasma is distinct from a gas because it possesses unique properties. Free electrical charges (not bound to atoms or ions) cause plasma to be electrically conductive. Plasma may be formed by heating and ionizing a gas

A comparison of the characteristic properties of solids, liquids and gases are given in Table 4.1.

Table 2.3.1. Comparison of Characteristic Properties of Solids, Liquids and Gases

PROPERTIES	SOLIDS	LIQUIDS	GAS
Shape of the container	Definite shape	Take the shape of the container	Take the shape of the container by occupying the whole space
Volume of the container	Definite	Definite	Take the volume
Compressibility	Almost nil	Almost nil	Very large

Fluidity or rigidity	Rigid	Fluid	Fluid
Density	Large	Large	Very small
Diffusion do not diffuse			

2.3.2 Why Solids, Liquids and Gases Exhibit Different Properties?

The properties of matter in the three states of matter are different because the characteristics of the particles vary in the three states of matter.

Now let us understand how the characteristics of particles vary in the three states of matter.

Changes in states

Chemical properties:

Properties that do change that chemical nature of matter

Examples of physical properties are: color, smell, freezing point, boiling point, melting point, infra-red spectrum, attraction (paramagnetic) or repulsion (diamagnetic) to magnets, opacity, viscosity and density. There are many more examples. Note that measuring each of these properties will not alter the basic nature of the substance.

Examples of chemical properties are: heat of combustion, reactivity with water, PH, and electromotive force.

The more properties we can identify for a substance, the better we know the nature of that substance. These properties can then help us model the substance and thus understand how this substance will behave under various conditions.

Physical and Chemical Properties

All substances have properties that we can use to identify them. For example we can identify a person by their face, their voice, height, finger prints, DNA etc.. The more of these properties that we can identify, the better we know the person. In a similar way matter has properties - and there are many of them. There are two basic types of properties that we can associate with matter. These properties are called Physical properties and Chemical properties:

Physical properties:

Properties that do not change the chemical nature of matter

Chemical properties:

Properties that do change chemical nature of matter

Examples of physical properties are: color, smell, freezing point, boiling point, melting point, infra-red spectrum, attraction (paramagnetic) or repulsion (diamagnetic) to magnets, opacity, viscosity and density. There are many more examples. Note that measuring each of these properties will not alter the basic nature of the substance.

Examples of chemical properties are: heat of combustion, reactivity with water, PH, and electromotive force.

The more properties we can identify for a substance, the better we know the nature of that substance. These properties can then help us model the substance and thus understand how this substance will behave under various conditions.

2.4 Changing States of Matter

A material will change from one state or phase to another at specific combinations of temperature and surrounding pressure. Typically, the pressure is atmospheric pressure, so temperature is the determining factor to the change in state in those cases.

Names such as boiling and freezing are given to the various changes in states of matter. The temperature of a material will increase until it reaches the point where the change takes place. It will stay at that temperature until that change is completed.

Changes in states

The states of matter are solid, liquid, gas and plasma. Since there is some debate on whether plasma should be classified as a state of matter and since it is not commonly experienced, we will not discuss its properties here.

Order of changes

When heat is applied to a material, its change in state typically goes from solid to liquid to gas. There are some exceptions where the material will go directly from a solid to a gas.

When a material is cooled, its change in state typically goes from gas to liquid to solid. There are some exceptions where the material will go directly from a gas to a solid.

Names of changes

Each change in the state of matter has a specific name.

Start from:	Change to:	Name
solid	liquid	melting
liquid	solid	freezing
liquid	gas	boiling

gas	liquid	condensation
solid	gas (skipping liquid phase)	sublimation
gas	solid (skipping liquid phase)	deposition

Change in temperature

When a material reaches the temperature at which a change in state occurs, the temperature will remain the same until all the energy is used to change the state.

Melting

When a solid is heated, its temperature rises until it reaches its melting point. Any additional heat added to the material will not raise the temperature until all of the material is melted.

Thus, if you heat some ice, its temperature will rise until it reaches 0° C (32° F). Then the ice will stay at that temperature until all the ice is melted. The heat energy is used to melt the ice and not to raise the temperature. After the ice is melted, the temperature of the water will continue to rise as more heat is applied.

Boiling

When a liquid is heated, its temperature rises until it reaches its boiling point. The temperature will then remain at that point until all of the liquid is boiled away.

For example, the temperature of a pot of water will increase until it reaches 100° C (212° F). It will stay there until all the water is boiled away. The temperature of the steam can then be increased.

Cooling

Likewise, when a gas is cooled, its temperature will drop until it reaches the condensation point. Any additional cooling or heat loss will not lower the temperature until all of the gas is condensed into the liquid state.

Then the temperature of the liquid will continue to drop as more cooling is applied. Once the liquid reaches the freezing point, the temperature will remain at that point until all of the liquid is solidified. Then the temperature of the solid can continue to decrease.

2.4.1 Chemical Changes

Chemical changes take place on the molecular level. A chemical change produces a new substance. Examples of chemical changes include combustion (burning), cooking an egg, rusting of an iron pan, and mixing hydrochloric acid and sodium hydroxide to make salt and water

2.4.2 Physical Changes

Physical changes are concerned with energy and states of matter.

A physical change does not produce a new substance. Changes in state or phase (melting, freezing, vaporization, condensation, sublimation) are physical changes. Examples of physical changes include crushing a can, melting an ice cube, and breaking a bottle.

2.5 How to Tell Chemical & Physical Changes Apart

A chemical change makes a substance that wasn't there before. There may be clues that a chemical reaction took place, such as light, heat, color change, gas production, odor, or sound. The starting and ending materials of a physical change are the same, even though they may look different.

Examples of Chemical Changes

- burning wood
- dissolving salt in water
- mixing acid and base
- digesting food

Examples of Physical Changes

- *crumpling a sheet of paper*
- *melting an ice cube*
- *casting silver in a mold*
- *breaking a bottle*

2.6 How to Tell?

Look for an indication that a chemical change occurred. Chemical reactions release or absorb heat or other energy or may produce a gas, odor, color or sound. If you don't see any of these indications, a physical change likely occurred.

In some cases, it may be hard to tell whether a chemical or physical change occurred. For example, when you dissolve sugar in water, a physical change occurs. The form of the sugar changes, but it remains the same chemically (sucrose molecules). However, when you dissolve salt in water the salt dissociates into its ions (from NaCl into Na⁺ and Cl⁻) so a chemical change occurs. In both cases a white solid dissolves into a clear liquid and in both cases you can recover the starting material by removing the water, yet the processes are not the same.

2.7 Difference between chemical and physical change

2.7.1 Chemical change is any change that results in the formation of new chemical substances. At the molecular level, chemical change involves making or breaking of bonds between atoms. These changes are chemical:

- Iron rusting (iron oxide forms)
- Gasoline burning (water vapor and carbon dioxide form)
- Eggs cooking (fluid protein molecules uncoil and crosslink to form a network)
- Bread rising (yeast converts carbohydrates into carbon dioxide gas)
- Milk souring (sour-tasting lactic acid is produced)
- Sun tanning (vitamin D and melanin is produced)

2.7.2 Physical change rearranges molecules but doesn't affect their internal structures. Some examples of physical change are:

- Whipping egg whites (air is forced into the fluid, but no new substance is produced)
- Magnetizing a compass needle (there is realignment of groups ("domains") of iron atoms, but no real change within the iron atoms themselves).
- Boiling water (water molecules are forced away from each other when the liquid changes to vapor, but the molecules are still H_2O .)
- Dissolving sugar in water (sugar molecules are dispersed within the water, but the individual sugar molecules are unchanged.) dicing potatoes (cutting usually separates molecules without changing them.)

SUMMARY

- Matter is anything that occupies space and has mass.
- Matter can be classified as solids, liquids and gases on the basis of its physical state.
- Matter is made up of extremely small particles.
- There are spaces between particles of matter.
- The particles of matter are continuously moving.
- The particles of matter attract each other.
- The spaces between particles are minimum in solid state and maximum in gases.
- The kinetic energy of particles minimum in solid state and maximum in gaseous state.
- The force of attraction between particles is maximum in solid state and negligible in gaseous state.
- Liquids and gases exhibit diffusion because their particles possess translatory motion and possess larger interparticle spaces.
- Solid and liquid states are known as condensed states of matter due to smaller interparticle spaces and very little compressibility.
- Liquids and gases are known as fluids because of their ability to flow and take the shape of the container.

EVALUATION

1. Which of the following is not an example of matter?

- (a) Air
- (b) Almonds
- (c) Cold-drink
- (d) Love.

2. Which of the following has the strongest interparticle forces?

- (a) Nitrogen
- (b) Water
- (c) Iron
- (d) Neon.

3 Which of the following has atoms as the constituent particles?

- (a) Dry ice
- (b) Argon
- (c) Glucose
- (d) Potassium chloride.

Fill in the blanks

Complete the following sentences by supplying appropriate words:

- (i) The particles in state do not possess translator motion.
- (ii) and states of matter are known as fluid states of matter.
- (iii) Particles in state possess maximum kinetic energy.
- (iv) Kinetic energy of particles of matter with increase in temperature.

STUDY SESSION THREE: ELEMENTS

3.0 Introduction

Chemistry is the branch of science that studies **matter**, its composition, structure, properties, and the changes it undergoes.

Chemistry is called the **central science** because it connects many areas of science, including physics, biology, medicine, environmental science, geology, and engineering.

This study session covers:

- The meaning and scope of chemistry.
- The major branches of chemistry.
- The classification and properties of matter.
- Applications of chemistry in everyday life, biology, the environment, and industry.
- The scientific method and chemical investigation.
- Basic chemistry laboratory equipment and their uses.
- Measurement techniques.
- Laboratory safety and proper handling of hazards.

3.1 Learning Outcomes

By the end of this study session, you should be able to:

1. Define chemistry and describe its scope as a scientific discipline.
2. Identify the major branches of chemistry, including:
 - Organic chemistry
 - Inorganic chemistry
 - Physical chemistry
 - Analytical chemistry
 - Biochemistry
3. Explain why chemistry is called the central science and how it connects different scientific disciplines.
4. Classify matter into elements, compounds, and mixtures.
5. Explain applications of chemistry in biological, environmental, and industrial processes.
6. Identify common laboratory equipment and explain their uses.
7. Describe basic laboratory safety procedures, including appropriate responses to chemical hazards.

3.2 ELEMENTS

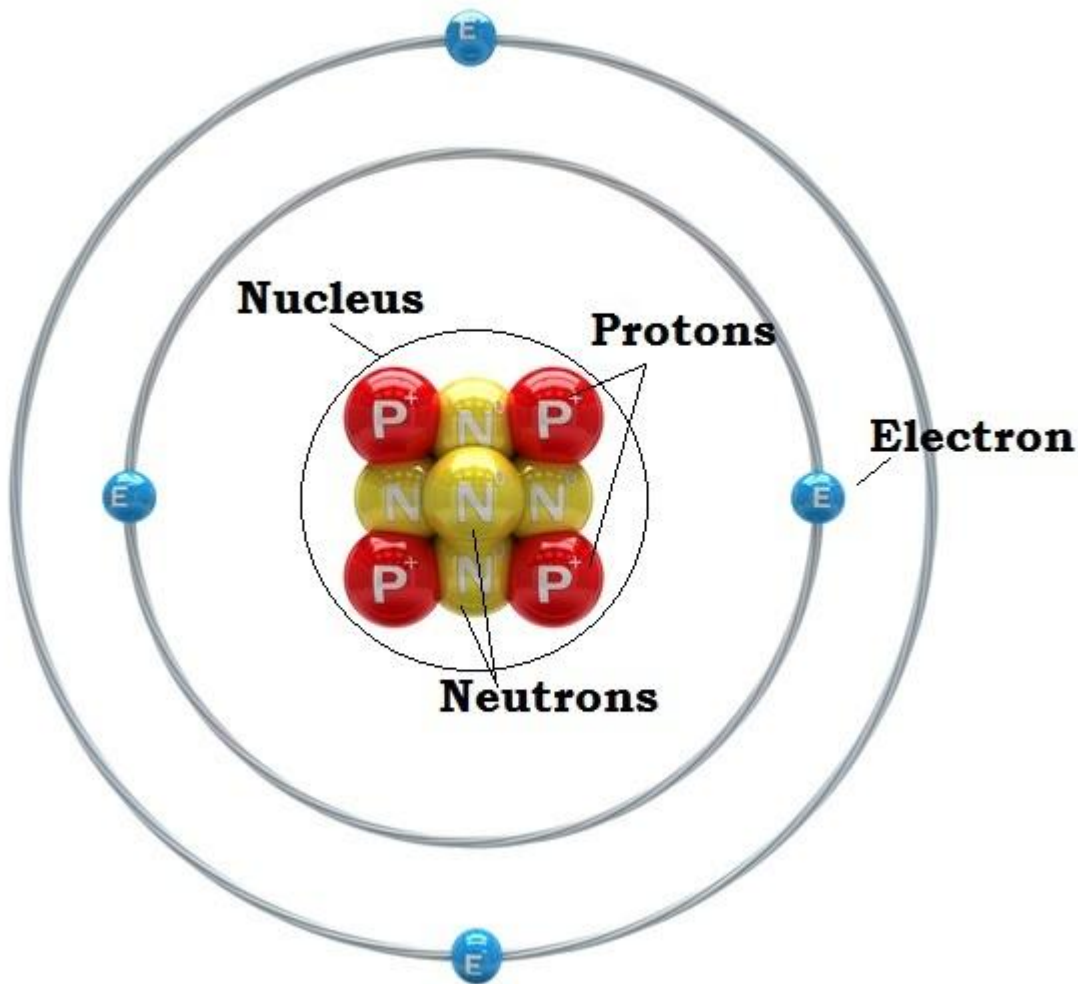
Element Definition: A chemical element is a substance that cannot be broken down by chemical means. Elements are defined by the number of protons they possess.

These are the first 20 elements, listed in order:

- 1 - H - Hydrogen
- 2 - He - Helium
- 3 - Li - Lithium
- 4 - Be - Beryllium
- 5 - B - Boron
- 6 - C - Carbon
- 7 - N - Nitrogen
- 8 - O - Oxygen
- 9 - F - Fluorine
- 10 - Ne - Neon
- 11 - Na - Sodium
- 12 - Mg - Magnesium
- 13 - Al - Aluminum
- 14 - Si - Silicon
- 15 - P - Phosphorus
- 16 - S - Sulfur
- 17 - Cl - Chlorine
- 18 - Ar - Argon
- 19 - K - Potassium
- 20 - Ca - Calcium

3.3 States of Matter - Elements

The states of matter of all of the elements is given for normal conditions, i.e. a temperature of 20°C. The 3 states of matter are solid, liquid or gas. Most elements are solids, only 11 are gases and 2 are liquids.



Atoms are the basic units of matter and the defining structure of elements. Atoms are made up of three particles: protons, neutrons and electrons.

Protons and neutrons are heavier than electrons and reside in the center of the atom, which is called the nucleus. Electrons are extremely lightweight and exist in a cloud orbiting the nucleus. The electron cloud has a radius 10,000 times greater than the nucleus.

Protons and neutrons have approximately the same mass. However, one proton weighs more than 1,800 electrons. Atoms always have an equal number of protons and electrons, and the number of protons and neutrons is usually the same as well. Adding a proton to an atom makes a new element, while adding a neutron makes an isotope, or heavier version, of that atom.

3.4 Nucleus

The nucleus was discovered in 1911, but its parts were not identified until 1932. Virtually all the mass of the atom resides in the nucleus. The nucleus is held together by the "strong force," one of the four basic forces in nature. This force between the protons and neutrons overcomes the repulsive electrical force that would, according to the rules of electricity, push the protons apart otherwise.

3.5 Protons

Protons are positively charged particles found within atomic nuclei. They were discovered by Ernest Rutherford in experiments conducted between 1911 and 1919.

The number of protons in an atom defines what element it is. For example, carbon atoms have six protons, hydrogen atoms have one and oxygen atoms have eight. The number of protons in an atom is referred to as the atomic number of that element. The number of protons in an atom also determines the chemical behavior of the element. The Periodic Table of the Elements arranges elements in order of increasing atomic number. Protons are made of other particles called quarks. There are three quarks in each proton — two "up" quarks and one "down" quark — and they are held together by other particles called gluons.

3.6 Electrons

Electrons have a negative charge and are electrically attracted to the positively charged protons. Electrons surround the atomic nucleus in pathways called orbitals. The inner orbitals surrounding the atom are spherical but the outer orbitals are much more complicated. An atom's electron configuration is the orbital description of the locations of the electrons in an unexcited atom. Using the electron configuration and principles of physics, chemists can predict an atom's properties, such as stability, boiling point and conductivity.

Typically, only the outermost electron shells matter in chemistry. The inner electron shell notation is often truncated by replacing the long-hand orbital description with the symbol for a noble gas in brackets. This method of notation vastly simplifies the description for large molecules.

For example, the electron configuration for beryllium (Be) is $1s^2 2s^2$, but it's written $[\text{He}]2s^2$. $[\text{He}]$ is equivalent to all the electron orbitals in a helium atom. The Letters, s, p, d, and f designate the shape of the orbitals and the superscript gives the number of electrons in that orbital.

3.7 Neutrons

Neutrons are uncharged particles found within atomic nuclei. A neutron's mass is slightly larger than that of a proton. Like protons, neutrons are also made of quarks — one "up" quark and two "down" quarks. Neutrons were discovered by James Chadwick in 1932.

The **mass number (A)**, also called **atomic mass number** or **nucleon number**, is the total number of protons and neutrons (together known as nucleons) in an atomic nucleus. It determines the atomic mass of atoms. Because protons and neutrons both are baryons, the mass number A is identical with the baryon number B as of the nucleus as of the whole atom or ion. The mass number is different for each different isotope of a chemical element. This is not the same as the atomic number (**Z**) which denotes the number of protons in a nucleus, and thus uniquely identifies an element. Hence, the difference between the mass number and the atomic number gives the number of neutrons (N) in a given nucleus: $N=A-Z$.^[1] The mass number is written either after the element name or as a superscript to the left of an element's symbol. For example, the most common isotope of carbon is carbon-12, or ^{12}C , which has 6 protons and 6 neutrons. The full isotope symbol would also have the atomic number (**Z**) as a subscript to the left of the element symbol directly below the mass number: $^{12}_{6}\text{C}$.^[2]

This is technically redundant, as each element is defined by its atomic number, so it is often omitted

The **atomic number** of a chemical element (also known as its **proton number**) is the number of protons found in the nucleus of an atom of that element, and therefore identical to the charge number of the nucleus. It is conventionally represented by the symbol **Z**. The atomic number uniquely identifies a chemical element. In an uncharged atom, the atomic number is also equal to the number of electrons

EVALUATION

1. Write the symbols of the first twenty elements.
2. Classify the first twenty elements into the three states of matter.
3. Define atomic number and mass number.

STUDY SESSION FOUR: MOLECULES AND ATOMICITY

INTRODUCTION

A molecule is the smallest particle in a chemical element or compound that has the chemical properties of that element or compound. Molecules are made up of atoms that are held together by chemical bonds. These bonds form as a result of the sharing or exchange of electrons among atoms. The atoms of certain elements readily bond with other atoms to form molecules. Examples of such elements are oxygen and chlorine. The atoms of some elements do not easily bond with other atoms. Examples are neon and argon.

Molecules can vary greatly in size and complexity. The element helium is a one-atom molecule. Some molecules consist of two atoms of the same element. For example, O_2 is the oxygen molecule most commonly found in the earth's atmosphere; it has two atoms of oxygen. However, under certain circumstances, oxygen atoms bond into triplets (O_3), forming a molecule known as ozone. Other familiar molecules include water, consisting of two hydrogen atoms and one oxygen atom (H_2O), carbon dioxide, consisting of one carbon atom bonded to two oxygen atoms (CO_2), and sulfuric acid, consisting of two hydrogen atoms, one sulfur atom, and four oxygen atoms (H_2SO_4).

4.1 ATOMICITY OF ELEMENTS.

1. MONOATOMIC ELEMENTS: these are elements with only one atom e.g. the noble gases (neon, argon, helium), sodium magnesium, in short all metals are monoatomic.

2. DIATOMIC ELEMENTS: these elements contain two atoms .most non metals fall under this category e.g. N_2 F_2 , Cl_2 , O_2 , Br_2 I_2 , H_2 .

3. POLYATOMIC ELEMENTS: are those elements with more than two atoms. e.g. phosphorus 5, sulphur 8

4.2 Foundations of Dalton's atomic theory

Dalton's atomic theory makes the following assumptions:

1. All matter consists of tiny particles. The existence of atoms was first suggested more than 2000 years before Dalton's birth. Atoms remained pure speculation through most of this time, although Newton used arguments based on atoms to explain the gas laws in 1687. (Newton's speculations about atoms in the *Principia* were carefully copied by hand into Dalton's notebooks.)

2. Atoms are indestructible and unchangeable. Atoms of an element cannot be created, destroyed, broken into smaller parts or transformed into atoms of another element. Dalton based this hypothesis on the law of conservation of mass and on centuries of experimental evidence.

With the discovery of subatomic particles after Dalton's time, it became apparent that atoms could be broken into smaller parts. The discovery of nuclear processes showed that it was even possible to transform atoms from one element into atoms of another. But we don't consider processes that affect the nucleus to be chemical processes. The postulate is still useful in explaining the law of conservation of mass in chemistry. A slightly more restrictive wording is "Atoms cannot be created, destroyed, or transformed into other atoms in a chemical change".

3. **Elements are characterized by the mass of their atoms.** All atoms of the same element have identical weights, Dalton asserted. Atoms of different elements have different weights. (Dalton used the word "weight" rather than mass, and chemists have called atomic masses "atomic weights" ever since).

We now know that atoms of the same element sometimes have slightly different masses, but always have identical nuclear charge. In modern atomic theory, the postulate has been amended to read: "Elements are characterized by the nuclear charge of their atoms".

4. **When elements react, their atoms combine in simple, whole-number ratios.** This postulate suggested a practical strategy for determining relative atomic weights from elemental percentages in compounds. Experimental atomic weights could then be used to explain the fixed mass percentages of elements in all compounds of those elements! By suggesting that compounds contained characteristic atom-to-atom ratios, Dalton effectively explained the law of definite proportions.

5. **When elements react, their atoms sometimes combine in more than one simple, whole-number ratio.** Dalton used this postulate to explain why the weight ratios of nitrogen to oxygen in various nitrogen oxides were themselves simple multiples of each other. Even Dalton's critics were impressed by the power and simplicity of his explanation, and it persuaded many of them that his atomic theory was worthy of further investigation.

Unfortunately, Dalton included an additional postulate that prevented his theory from being accepted for many years. When atoms combine in only one ratio, Dalton said, "...it must be presumed to be a *binary* one, unless some cause appear to the contrary" [2]. He had no experimental evidence to support this postulate, and it led him to mistakenly assume that the formula of water was OH and the formula of ammonia was NH. As a result, Dalton's atomic weights for oxygen and nitrogen were incorrect and his experimental data did not support many of the conclusions he drew from it. A consistent set of atomic weights was absolutely essential before the theory could be accepted and applied. Next, we'll see how Dalton's postulates can be used to estimate atomic weights from experimental data, and how they explain three basic laws of chemistry.

EVALUATION

1. Define molecules and Atomicity.
2. State with examples, the types of Atomicity.
3. State Dalton's Atomic theory. Describe its modification

STUDY SESSION FIVE: PARTICULATE NATURE OF MATTER

5.0 INTRODUCTION:

Matter is made up of **tiny particles** that are too small to be seen with the naked eye. These particles may be atoms, molecules, or ions, and their arrangement and movement determine the properties of matter.

The **particulate nature of matter** explains that all matter is composed of very small particles that are constantly moving and have spaces between them.

This study session covers:

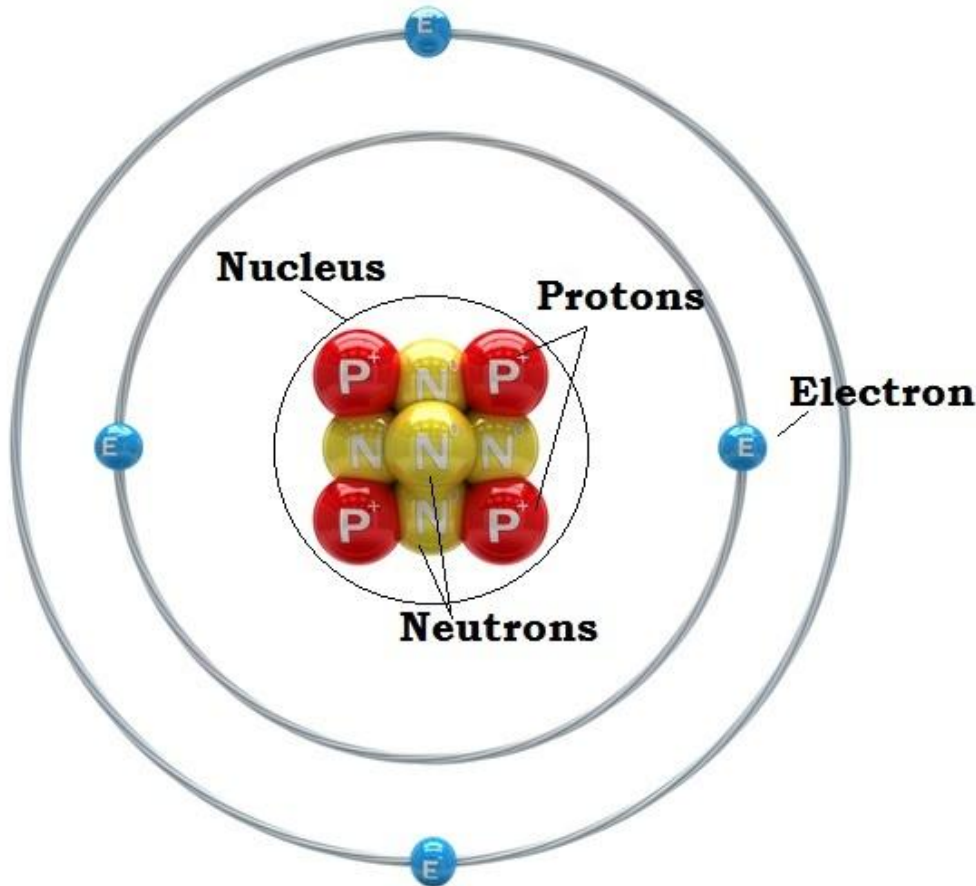
- The meaning of the particulate nature of matter.
- The main ideas of the **particle theory of matter**.
- The arrangement and movement of particles in solids, liquids, and gases.
- The spaces between particles.
- The forces of attraction between particles.
- The effect of temperature on particle movement.
- Diffusion and Brownian motion as evidence for the particulate nature of matter.

Learning Outcomes

By the end of this study session, you should be able to:

1. **Explain the particulate nature of matter and state that matter is made up of tiny particles.**
2. **Describe the particle theory of matter, including:**
 - Matter is made up of tiny particles.
 - Particles are in constant motion.
 - There are spaces between particles.
 - Particles attract one another.
 - Particles move faster when temperature increases.
3. **Describe the arrangement of particles in:**
 - Solids
 - Liquids
 - Gases
4. **Compare the movement of particles in solids, liquids, and gases.**
5. **Explain how temperature affects** particle movement and the kinetic energy of particles.
6. **Explain diffusion as the movement of particles** from a region of higher concentration to a region of lower concentration.
7. Describe Brownian motion as evidence that particles of matter are constantly moving.
8. Relate the particulate nature of matter to changes of state, including melting, freezing, evaporation, boiling, condensation, and sublimation.

Fig5.1



Atoms are the basic units of matter and the defining structure of elements. Atoms are made up of three particles: protons, neutrons and electrons.

Protons and neutrons are heavier than electrons and reside in the center of the atom, which is called the nucleus. Electrons are extremely lightweight and exist in a cloud orbiting the nucleus. The electron cloud has a radius 10,000 times greater than the nucleus.

Protons and neutrons have approximately the same mass. However, one proton weighs more than 1,800 electrons. Atoms always have an equal number of protons and electrons, and the number of protons and neutrons is usually the same as well. Adding a proton to an atom makes a new element, while adding a neutron makes an isotope, or heavier version, of that atom.

5.2 Nucleus

The nucleus was discovered in 1911, but its parts were not identified until 1932. Virtually all the mass of the atom resides in the nucleus. The nucleus is held together by the "strong force," one of the four basic forces in nature. This force between the protons and neutrons overcomes the repulsive electrical force that would, according to the rules of electricity, push the protons apart otherwise.

5.3 Protons

Protons are positively charged particles found within atomic nuclei. They were discovered by Ernest Rutherford in experiments conducted between 1911 and 1919.

The number of protons in an atom defines what element it is. For example, carbon atoms have six protons, hydrogen atoms have one and oxygen atoms have eight. The number of protons in an atom is referred to as the atomic number of that element. The number of protons in an atom also determines the chemical behavior of the element. The Periodic Table of the Elements arranges elements in order of increasing atomic number.

Protons are made of other particles called quarks. There are three quarks in each proton — two "up" quarks and one "down" quark — and they are held together by other particles called gluons.

5.4 Electrons

Electrons have a negative charge and are electrically attracted to the positively charged protons. Electrons surround the atomic nucleus in pathways called orbitals. The inner orbitals surrounding the atom are spherical but the outer orbitals are much more complicated. An atom's electron configuration is the orbital description of the locations of the electrons in an unexcited atom. Using the electron configuration and principles of physics, chemists can predict an atom's properties, such as stability, boiling point and conductivity.

Typically, only the outermost electron shells matter in chemistry. The inner electron shell notation is often truncated by replacing the long-hand orbital description with the symbol for a noble gas in brackets. This method of notation vastly simplifies the description for large molecules.

For example, the electron configuration for beryllium (Be) is $1s^2 2s^2$, but it's written $[\text{He}]2s^2$. $[\text{He}]$ is equivalent to all the electron orbitals in a helium atom. The letters, s, p, d, and f designate the shape of the orbitals and the superscript gives the number of electrons in that orbital.

5.5 Neutrons

Neutrons are uncharged particles found within atomic nuclei. A neutron's mass is slightly larger than that of a proton. Like protons, neutrons are also made of quarks — one "up" quark and two "down" quarks. Neutrons were discovered by James Chadwick in 1932.

5.6 RULES FOR FILLING ORBITALS

Rule 1 - Lowest energy orbitals fill first. Thus, the filling pattern is **1s, 2s, 2p, 3s, 3p, 4s, 3d**, etc. Since the orbitals within a subshell are degenerate (of equal energy), the entire subshell of a particular orbital type is filled before moving to the next subshell of higher energy.

Rule 2 - Pauli Exclusion Principle - Only two electrons are permitted per orbital and they must be of opposite spin. If one electron within an orbital possesses a clockwise spin, then the second electron within that orbital will possess a counterclockwise spin. Two electrons with opposite

spins found in the same orbital are referred to as being paired.

Rule 3- Hund's Rule - The most stable arrangement of electrons in a subshell occurs when the maximum number of unpaired electrons exists, all possessing the same spin direction. This occurs due to the degeneracy of the orbitals, all orbitals within a subshell are of equal energy. Electrons are repulsive to one another and only pair after all of the orbitals have been singly filled.

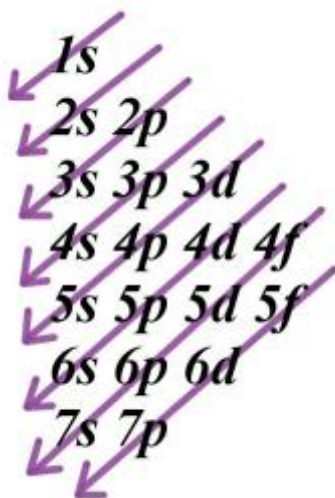
Rules for Assigning Electron Orbitals

5.6.1 Occupation of Orbitals

Electrons fill orbitals in a way to minimize the energy of the atom. Therefore, the electrons in an atom fill the principal energy levels in order of increasing energy (the electrons are getting farther from the nucleus). The order of levels filled looks like this:

1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s, 5f, 6d, and 7p

One way to remember this pattern, probably the easiest, is to refer to the periodic table and remember where each orbital block falls to logically deduce this pattern. Another way is to make a table like the one below and use vertical lines to determine which subshells correspond with each other.



5.6.2 Pauli Exclusion Principle

The Pauli exclusion principle states that no two electrons can have the same four quantum numbers. The first three (n , l , and m_l) may be the same, but the fourth quantum number must be different. A single orbital can hold a maximum of two electrons, which **must** have opposing spins; otherwise they would have the same four quantum numbers, which is forbidden. One

electron is spin up ($m_s = +1/2$) and the other would spin down ($m_s = -1/2$). This tells us that each subshell has double the electrons per orbital. The s subshell has 1 orbital that can hold up to 2 electrons, the p subshell has 3 orbitals that can hold up to 6 electrons, the d subshell has 5 orbitals that hold up to 10 electrons, and the f subshell has 7 orbitals with 14 electrons.

Example 1: Hydrogen and Helium

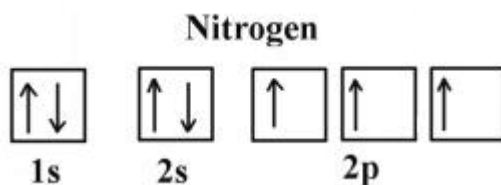
The first three quantum numbers of an electron are $n=1$, $l=0$, $m_l=0$. Only two electrons can correspond to these, which would be either $m_s = -1/2$ or $m_s = +1/2$. As we already know from our studies of quantum numbers and electron orbitals, we can conclude that these four quantum numbers refer to the 1s subshell. If only one of the m_s values are given then we would have $1s^1$ (denoting hydrogen) if both are given we would have $1s^2$ (denoting helium). Visually, this is represented as:

5.6.3 Hund's Rule

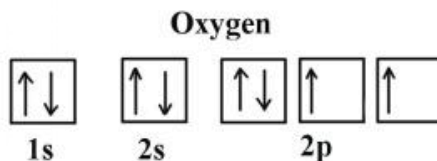
When assigning electrons in orbitals, each electron will first fill all the orbitals with similar energy (also referred to as degenerate) before pairing with another electron in a half-filled orbital. Atoms at ground states tend to have as many unpaired electrons as possible. When visualizing these processes, think about how electrons are exhibiting the same behavior as the same poles on a magnet would if they came into contact; as the negatively charged electrons fill orbitals they first try to get as far as possible from each other before having to pair up.

Example 2: Oxygen and Nitrogen

If we look at the correct electron configuration of the Nitrogen ($Z = 7$) atom, a very important element in the biology of plants: $1s^2 2s^2 2p^3$



We can clearly see that p orbitals are half-filled as there are three electrons and three p orbitals. This is because Hund's Rule states that the three electrons in the 2p subshell will fill all the empty orbitals first before filling orbitals with electrons in them. If we look at the element after Nitrogen in the same period, Oxygen ($Z = 8$) its electron configuration is: $1s^2 2s^2 2p^4$ (for an atom).



Oxygen has one more electron than Nitrogen and as the orbital are all half filled the electron must pair up.

5.7 The Aufbau Process

Aufbau comes from the German word "aufbauen" meaning "to build." When writing electron configurations, orbitals are built up from atom to atom. When writing the electron configuration for an atom, orbitals are filled in order of increasing atomic number. However, there are some exceptions to this rule.

Example 3: 3rd row elements

Following the pattern across a period from B (Z=5) to Ne (Z=10), the number of electrons increases and the subshells are filled. This example focuses on the p subshell, which fills from boron to neon.

- B (Z=5) configuration: $1s^2 2s^2 2p^1$
- C (Z=6) configuration: $1s^2 2s^2 2p^2$
- N (Z=7) configuration: $1s^2 2s^2 2p^3$
- O (Z=8) configuration: $1s^2 2s^2 2p^4$
- F (Z=9) configuration: $1s^2 2s^2 2p^5$
- Ne (Z=10) configuration: $1s^2 2s^2 2p^6$

Exceptions

Although the Aufbau rule accurately predicts the electron configuration of most elements, there are notable exceptions among the transition metals and heavier elements. The reason these exceptions occur is that some elements are more stable with fewer electrons in some subshells and more electrons in others (Table 1).

Table 1: Exceptions to Electron Configuration Trends

Period 4:	Period 5:
Chromium: Z:24 [Ar] $3d^5 4s^1$	Niobium: Z:41 [Kr] $5s^1 4d^4$
Copper: Z:29 [Ar] $3d^{10} 4s^1$	Molybdenum: Z:42 [Kr] $5s^1 4d^5$
	Ruthenium: Z:44 [Kr] $5s^1 4d^7$
	Rhodium: Z:45 [Kr] $5s^1 4d^8$
	Palladium: Z:46 [Kr] $4d^{10}$
	Silver: Z:47 [Kr] $5s^1 4d^{10}$
Period 6:	Period 7:
Lanthanum: Z:57 [Xe] $6s^2 5d^1$	Actinium: Z:89 [Rn] $7s^2 6d^1$
Cerium: Z:58 [Xe] $6s^2 4f^1 5d^1$	Thorium: Z:90 [Rn] $7s^2 6d^2$
Gadolinium: Z:64 [Xe] $6s^2 4f^7 5d^1$	Protactinium: Z:91 [Rn] $7s^2 5f^2 6d^1$
Platinum: Z:78 [Xe] $6s^1 4f^{14} 5d^9$	Uranium: Z:92 [Rn] $7s^2 5f^3 6d^1$
Gold: Z:79 [Xe] $6s^1 4f^{14} 5d^{10}$	Neptunium: Z:93 [Rn] $7s^2 5f^4 6d^1$
	Curium: Z:96 [Rn] $7s^2 5f^7 6d^1$
	Lawrencium: Z:103 [Rn] $7s^2 5f^{14} 7p^1$

5.8 Writing Electron Configurations

When writing an electron configuration, first write the **energy level** (the period), then the **subshell** to be filled and the **superscript**, which is the number of electrons in that subshell. The

total number of electrons is the atomic number, Z . The rules above allow one to write the electron configurations for all the elements in the periodic table.

Three methods are used to write electron configurations:

1. Orbital diagrams
2. spdf notation
3. Noble Gas notation

Each method has its own purpose and each has its own drawbacks.

5.9 Orbital Diagrams

An orbital diagram, like those shown above, is a visual way to reconstruct the electron configuration by showing each of the separate orbitals and the spins on the electrons. This is done by first determining the subshell (s,p,d, or f) then drawing in each electron according to the stated rules above.

Example 4: Aluminum and Iridium

Write the electron configuration for aluminum and iridium.

SOLUTION

Aluminum is in the 3rd period and it has an atomic number of $Z=13$. If we look at the periodic table we can see that it is in the p-block as it is in group 13. Now we shall look at the orbitals it will fill: 1s, 2s, 2p, 3s, 3p. We know that aluminum completely fills the 1s, 2s, 2p, and 3s orbitals because mathematically this would be $2+2+6+2=12$. The last electron is in the 3p orbital. Also another way of thinking about it is that as you move from each orbital block, the subshells become filled as you complete each section of the orbital in the period. The block that the atom is in (in the case for aluminum: 3p) is where we will count to get the number of electrons in the last subshell (for aluminum this would be one electron because it's the first element in the period 3 p-block). This gives the following:

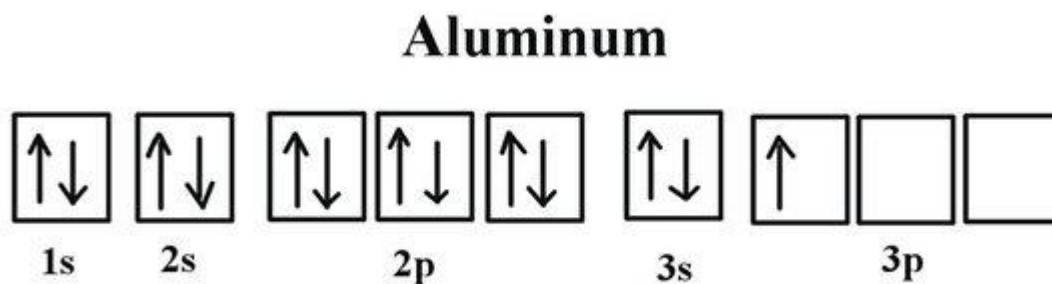
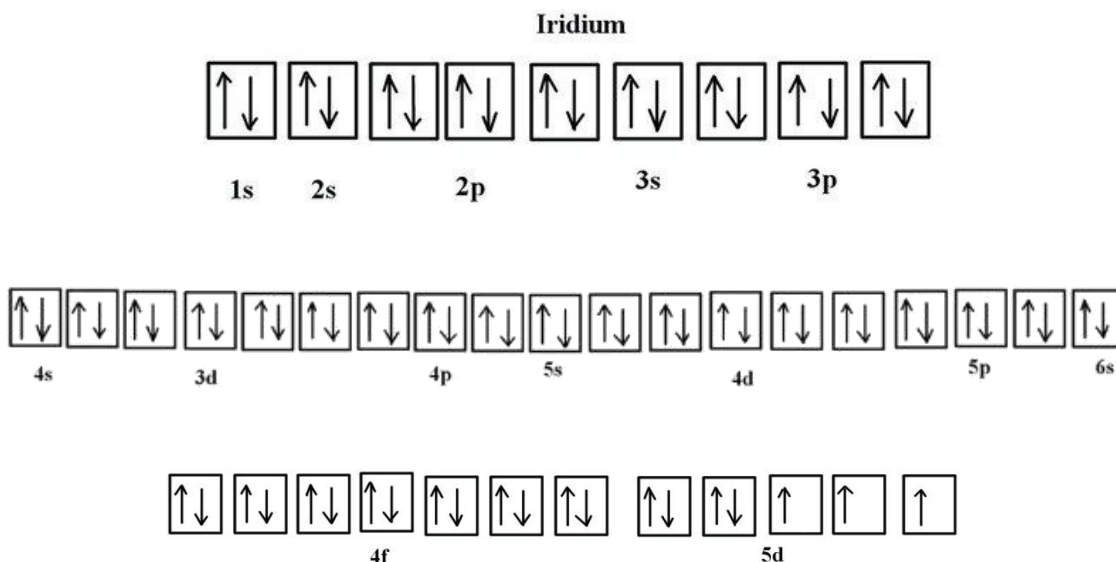


Fig5.9.1

Note that in the orbital diagram, the two opposing spins of the electron can be visualized. This is why it is sometimes useful to think about electron configuration in terms of the diagram. However, because it is the most time consuming method, it is more common to write or see electron configurations in spdf notation and noble gas notation. Another example is the electron

configuration of iridium:



The electron configuration of iridium is much longer than aluminum. Although drawing out each orbital may prove to be helpful in determining unpaired electrons, it is very time consuming and often not as practical as the spdf notation, especially for atoms with much longer configurations. Hund's rule is also followed, as each electron fills up each 5d orbital before being forced to pair with another electron.

spdf Notation

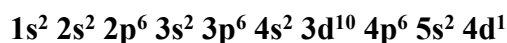
The most common way to describe electron configurations is to write distributions in the spdf notation. Although the distributions of electrons in each orbital are not as apparent as in the diagram, the total number of electrons in each energy level is described by a superscript that follows the relating energy level. To write the electron configuration of an atom, identify the energy level of interest and write the number of electrons in the energy level as its superscript as follows: $1s^2$. This is the electron configuration of helium; it denotes a full s orbital. The periodic table is used as a reference to accurately write the electron configurations of all atoms.

Example 5: Yttrium

Write the electronic configuration of Yttrium.

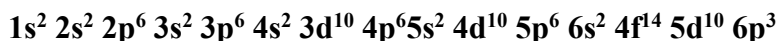
SOLUTION

Start with the straightforward problem of finding the electron configuration of the element yttrium. As always, refer to the periodic table. The element yttrium (symbolized Y) is a transition metal, found in the fifth period and in Group 3. In total it has thirty-nine electrons. Its electron configuration is as follows:



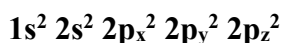
This is a much simpler and more efficient way to portray electron configuration of an atom. A logical way of thinking about it is that all that is required is to fill orbitals across a period and through orbital blocks. The number of elements in each block is the same as in the energy level it corresponds. For example, there are 2 elements in the s-block, and 10 elements in the d-block. Moving across, simply count how many elements fall in each block. Yttrium is the first element in the fourth period d-block; thus there is one electron in that energy level. To check the answer,

verify that the subscripts add up to the atomic number. In this case, $2+2+6+2+6+2+10+6+2+1=39$ and $Z=39$, so the answer is correct. A slightly more complicated example is the electron configuration of bismuth (symbolized Bi, with $Z = 83$). The periodic table gives the following electron configuration:

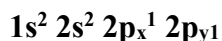


The reason why this electron configuration seems more complex is that the f-block, the Lanthanide series, is involved. Most students who first learn electron configurations often have trouble with configurations that must pass through the f-block because they often overlook this break in the table and skip that energy level. It's important to remember that when passing the 5d and 6d energy levels that one must pass through the f-block lanthanoid and actinoid series. Keeping this in mind, this "complex" problem is greatly simplified.

Another method (but less commonly used) of writing the spdf notation is the expanded notation format. This is the same concept as before, except that each individual orbital is represented with a subscript. The p, d, and f orbitals have different sublevels. The p orbitals are p_x, p_y , and p_z , and if represented on the 2p energy with full orbitals would look like: $2p_x^2 2p_y^2 2p_z^2$. The expanded notation for neon (Ne, $Z=10$) is written as follows:



The individual orbitals are represented, but the spins on the electrons are not; opposite spins are assumed. When representing the configuration of an atom with half filled orbitals, indicate the two half filled orbitals. The expanded notation for carbon is written as follows:



Because this form of the spdf notation is not typically used, it is not as important to dwell on this detail as it is to understand how to use the general spdf notation.

Noble Gas Notation

This brings up an interesting point about elements and electron configurations. As the p subshell is filled in the above example about the Aufbau principle (the trend from boron to neon), it reaches the group commonly known as the noble gases. The noble gases have the most stable electron configurations, and are known for being relatively inert. All noble gases have their subshells filled and can be used them as a shorthand way of writing electron configurations for subsequent atoms. This method of writing configurations is called the noble gas notation, in which the noble gas in the period above the element that is being analyzed is used to denote the subshells that element has filled and after which the valence electrons (electrons filling orbitals in the outer most shells) are written.

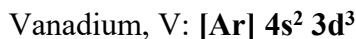
This looks slightly different from spdf notation, as the reference noble gas must be indicated.

Example 6: Vanadium

What is the electronic configuration of vanadium (V, $Z=23$)?

SOLUTION

Vanadium is the transition metal in the fourth period and the fifth group. The noble gas preceding it is argon (Ar, $Z=18$), and knowing that vanadium has filled those orbitals before it, argon is used as the reference noble gas. The noble gas in the configuration is denoted E, in brackets: [E]. To find the valence electrons that follow, subtract the atomic numbers: $23 - 18 = 5$. Instead of 23 electrons to distribute in orbitals, there are 5. Now there is enough information to write the electron configuration:



This method streamlines the process of distributing electrons by showing the valence electrons, which determine the chemical properties of atoms. In addition, when determining the number of unpaired electrons in an atom, this method allows quick visualization of the configurations of the valence electrons. In the example above, there are a full s orbital and three half filled d orbitals.

5.9B Ions: Atoms with an Electrical Charge

Atoms (or groups of atoms) in which there are unequal numbers of protons and electrons are called *ions*. Usually, the number of protons and electrons in atoms are equal. But there are cases in which an atom can acquire an electrical charge.

An ion example

For example, in the compound sodium chloride — table salt — the sodium atom has a positive charge and the chlorine atom has a negative charge.

The neutral sodium atom has 11 protons and 11 electrons, which means it has 11 positive charges and 11 negative charges. Overall, the sodium atom is neutral, and it's represented like this: Na. But the sodium ion contains one more positive charge than negative charge, so it's represented like this: Na^+

This unequal number of negative and positive charges can occur in one of two ways: An atom can gain a proton (a positive charge) or lose an electron (a negative charge).

Cations and anions

So which process is more likely to occur? In general, it's easy to gain or lose electrons but very difficult to gain or lose protons. So atoms become ions by gaining or losing electrons. And ions that have a positive charge are called *cations*.

The progression goes like this: The sodium ion shown above is formed from the loss of one electron. Because it lost an electron, it has more protons than electrons, or more positive charges than negative charges, which means it's now called the:

Na^+ cation

Likewise, when the neutral magnesium atom loses two electrons, it forms the:

Mg^{2+} cation

Now consider the chlorine atom in sodium chloride. The neutral chlorine atom has acquired a negative charge by gaining an electron. Because it has unequal numbers of protons and electrons, it's now an ion. And because ions that have a negative charge are called *anions*, it's now called the:

Cl^- anion

Other details about ions

Here are some extra tidbits about ions:

- You can write electron configurations and energy level diagrams for ions. The neutral sodium atom (11 protons) has an electron configuration of:
 $1s^2 2s^2 2p^6 3s^1$
The sodium cation has lost an electron — the valence electron, which is farthest away from the nucleus (the 3s electron, in this case). The electron configuration of the sodium ion is:
 $1s^2 2s^2 2p^6$
- The electron configuration of the chloride ion is:
 $1s^2 2s^2 2p^6 3s^2 3p^6$
This is the same electron configuration as the neutral Argon atom. If two chemical species have the same electron configuration, they're said to be *isoelectronic*.
- The preceding examples are all monoatomic (one atom) ions. But polyatomic (many atom) ions do exist. The ammonium ion is a polyatomic ion, or, specifically, a polyatomic cation. It is written as:
 NH_4^+
The nitrate ion, is also a polyatomic ion, or, specifically, a polyatomic anion. It is written as
 NO_3^-
- Ions are commonly found in a class of compounds called salts, or ionic solids. Salts, when melted or dissolved in water, yield solutions that conduct electricity. A substance that conducts electricity when melted or dissolved in water is called an *electrolyte*. Table salt — sodium chloride — is a good example.

On the other hand, when table sugar (sucrose) is dissolved in water, it becomes a solution that doesn't conduct electricity. So sucrose is a non-electrolyte.

Whether a substance is an electrolyte or a nonelectrolyte gives clues to the type of bonding in

the compound. If the substance is an electrolyte, the compound is probably ionically bonded. If it's a nonelectrolyte, it's probably covalently bonded.

EVALUATION

1. Write the electronic configuration of the first twenty elements.
2. Write out the characteristics of the first three fundamental particles in an atom.
3. An atom of an element is represented by X. How many electrons, protons and neutrons are in the atom? Write the electronic structure of the atom.

STUDY SESSION SIX: THE RELATIVE ATOMIC MASSES OF ELEMENTS.

6.0 INTRODUCTION

The **relative atomic mass** of an element is the average mass of its atoms compared with **one-twelfth of the mass of a carbon-12 atom**. It is represented by A_r and has no unit.

The relative atomic mass is important in chemistry because it helps us compare the masses of atoms of different elements and is used in calculating **relative molecular masses, formula masses, and chemical quantities**.

This study session covers:

- The meaning of relative atomic mass.
- The standard used for measuring atomic masses.
- The relative atomic mass scale.
- Isotopes and their effect on relative atomic mass.
- How to calculate the relative atomic mass of an element.
- The difference between relative atomic mass and mass number.
- The importance of relative atomic masses in chemical calculations.

Learning Outcomes:

By the end of this study session, you should be able to:

- Define relative atomic mass and state its symbol, A_r .
- Explain the carbon-12 standard used in determining relative atomic masses.
- Distinguish between relative atomic mass and mass number.
- Explain isotopes and describe how they affect the relative atomic mass of an element.
- Calculate the relative atomic mass of an element from the masses and abundances of its isotopes.
- Use relative atomic masses to calculate relative molecular mass and formula mass.
- Interpret relative atomic mass values from the periodic table.

6.1 ISOTOPE

Isotopes Definition: Isotopes are atoms with the same number of protons, but differing numbers of neutrons. Isotopes are different forms of a single element.

Examples: Carbon 12 and Carbon 14 are both Isotopes of carbon, one with 6 neutrons and one with 8 neutrons (both with 6 protons).

6.2 ATOMIC WEIGHTS AND ISOTOPIC ABUNDANCE

The atomic weight of an element is the relative atomic mass of that element. It is actually a weighted mass of the elements isotopes (if any) and their relative abundance.

You know that the sum of the percentages of the isotopes is equal to 1 (100%), so the relative abundance of the isotopes can be found using simple algebra.

Example 6.2.1:

Silver (Atomic weight 107.868) has two naturally-occurring isotopes with isotopic weights of 106.90509 and 108.90470. What is the percentage abundance of the lighter isotope?

To avoid mistakes, use "x" as the multiplier for the isotope percentage you wish to find. In this case, you want to find the percentage of the lighter isotope, so the "x" is associated with 106.90509. Since the sum of the isotopic abundance percentages is equal to 1 (100%), the formula is:

$$108.90470 (1 - x) + 106.90509 (x) = 107.868$$

Multiplying, re-arranging and condensing the above formula results in:

$$\begin{aligned} 108.90470 - 108.90470x + 106.90509x &= 107.868 \\ - 108.90470x + 106.90509x &= - 108.90470 + 107.868 \end{aligned}$$

$$- 1.9996x = - 1.0367$$

$$x = 0.5185$$

Therefore, the answer is 51.85 %

Example 6.2.2:

An imaginary element (Atomic weight 93.7140) has three naturally-occurring isotopes with isotopic weights of 92.9469, 93.2923 and 94.9030. The abundance of the lightest isotope is 42.38 %. What is the percentage abundance of the heaviest isotope?

In this case, we know the abundance of one of the isotopes. We know the percentages of the lighter isotope (42.38 %) and the percentage of the heavier isotope (x), so the percentage of the middle isotope is equal to 1 (100%) minus the other two percentages (1 - 0.4238 - x).

$$\begin{aligned} 92.9469 (0.4238) + 93.2923 [(1-0.4238)-x] + 94.9030x &= 93.7140 \\ 39.3909 + 53.7550 - 93.2923x + 94.9030x &= 93.7140 \end{aligned}$$

$$93.14 + 1.6107x = 93.7140$$

$$1.6107x = 0.4217$$

$$x = 26.18$$

Therefore, the answer is 26.18%

EVALUATION

1. Define the following terms I. Atomic number ii. mass number iii. Isotopes iv. Isotopy
2. Determine the relative atomic mass of carbon from a sample with the following data. 98.9% of carbon -12 and 1.1% of carbon-13.

STUDY SESSION SEVEN: COMPOUND

7.0 INTRODUCTION

A **compound** is a pure substance formed when **two or more different elements chemically combine** in fixed proportions. The elements in a compound are held together by **chemical bonds** and the compound has properties that are different from those of the elements that form it.

For example, **water (H₂O)** is a compound made up of hydrogen and oxygen chemically combined in a fixed ratio of 2:1.

This study session covers:

- The meaning of a compound.
- How compounds are formed.
- The difference between elements and compounds.
- The difference between compounds and mixtures.
- The properties of compounds.
- Chemical formulae and symbols of compounds.
- Examples of common compounds and their uses.

Learning Outcomes:

By the end of this study session, you should be able to:

1. Define a compound and explain how it is formed.
2. Explain chemical combination and the role of chemical bonds in forming compounds.
3. Identify the elements present in a given compound.
4. Write and interpret chemical formulae of simple compounds.
5. Distinguish between compounds and mixtures.
6. State the properties of compounds, including:
 - They contain two or more different elements.
 - The elements are chemically combined.
 - They have fixed composition.
 - They have properties different from their constituent elements.
7. Give examples of common compounds, such as water (H₂O), carbon dioxide (CO₂), sodium chloride (NaCl), and ammonia (NH₃).

7.1 COMPOUND

In chemistry, a compound is a substance that results from a combination of two or more different chemical elements, in such a way that the atoms of the different elements are held together by chemical bonds that are difficult to break. These bonds form as a result of the sharing or exchange of electrons among the atoms. The smallest unbreakable unit of a compound is called a molecule

Examples of compounds:

- water (H₂O)
- table salt (NaCl)
- sucrose (table sugar, C₁₂H₂₂O₁₁)

The relationship is simple.

7.2 Atoms, Elements, Molecules, and Compounds

7.2.1 Atoms are the basic building blocks of matter. An atom is the smallest unit of an element that retains the chemical properties of that element.

7.2.2 Elements are pure substances made up of only one type of atom. Each element has its own unique chemical properties. For example, **aluminum** consists entirely of aluminum atoms and has properties that distinguish it from other elements.

7.2.3 Molecules are formed when two or more atoms are chemically joined together. The atoms in a molecule may be from the same element or from different elements. For example, an oxygen molecule (O_2) contains two oxygen atoms, while a water molecule (H_2O) contains hydrogen and oxygen atoms.

7.2.4 Compounds are pure substances formed when atoms of **two or more different elements** chemically combine in fixed proportions. A compound has properties that are different from those of the elements from which it is formed. For example, water is a compound formed from hydrogen and oxygen.

Long before the chemical formulas of compounds were fully understood, chemists developed a system of **nomenclature**, which is the systematic naming of chemical substances. Today, chemical formulas such as NaCl , $\text{C}_{12}\text{H}_{22}\text{O}_{11}$, and $\text{Co}(\text{NH}_3)_6(\text{ClO}_4)_3$ are commonly used to represent compounds. However, chemical names are still important because they provide a standard way of identifying and communicating about chemical substances unique names that unambiguously identify each compound.

7.3 Common Names

Some compounds have been known for so long that a systematic nomenclature cannot compete with well-established common names. Examples of compounds for which common names are used include water (H_2O), ammonia (NH_3), and methane (CH_4).

7.3.1 Naming Ionic Compounds (Metals with Non-metals)

The names of ionic compounds are written by listing the name of the positive ion followed by the name of the negative ion.

NaCl	sodium chloride
$(\text{NH}_4)_2\text{SO}_4$	Ammonium sulfate
NaHCO_3	sodium bicarbonate

We therefore need a series of rules that allow us to unambiguously name positive and negative ions before we can name the salts these ions form.

7.3.2 Naming Positive Ions

Monatomic positive ions have the name of the element from which they are formed.

Na^+	sodium	Zn^{2+}	zinc
Ca^{2+}	calcium	H^+	hydrogen
K^+	potassium	Sr^{2+}	strontium

Some metals form positive ions in more than one oxidation state. One of the earliest methods of distinguishing between these ions used the suffixes *-ous* and *-ic* added to the Latin name of the element to represent the lower and higher oxidation states, respectively.

Fe^{2+}	ferrous	Fe^{3+}	ferric
Sn^{2+}	stannous	Sn^{4+}	stannic
Cu^+	cuprous	Cu^{2+}	cupric

Chemists now use a simpler method, in which the charge on the ion is indicated by a Roman numeral in parentheses immediately after the name of the element.

Fe^{2+}	iron(II)	Fe^{3+}	iron (III)
Sn^{2+}	tin(II)	Sn^{4+}	tin(IV)
Cu^+	copper(I)	Cu^{2+}	copper(II)

Polyatomic positive ions often have common names ending with the suffix *-onium*.

H_3O^+	hydronium
NH_4^+	ammonium

7.3.3 Naming Negative Ions

Negative ions that consist of a single atom are named by adding the suffix *-ide* to the stem of the name of the element.

F^-	fluoride	O^{2-}	oxide
Cl^-	chloride	S^{2-}	sulfide
Br^-	bromide	N^{3-}	nitride
I^-	iodide	P^{3-}	phosphide
H^-	hydride	C^{4-}	carbide

7.3.4 Common Polyatomic Negative Ions

-1 ions

HCO_3^-	bicarbonate	HSO_4^-	hydrogen sulfate (bisulfate)
CH_3CO_2^-	acetate	ClO_4^-	perchlorate
NO_3^-	nitrate	ClO_3^-	chlorate
NO_2^-	nitrite	ClO_2^-	chlorite
MnO_4^-	permanganate		Hypochlorite
CN^-	cyanide	ClO^-	
		OH^-	hydroxide

-2 ions

CO_3^{2-}	carbonate	O_2^{2-}	peroxide
SO_4^{2-}	sulfate	CrO_4^{2-}	chromate
SO_3^{2-}	sulfite	$\text{Cr}_2\text{O}_7^{2-}$	dichromate
$\text{S}_2\text{O}_3^{2-}$	thiosulfate	HPO_4^{2-}	hydrogen phosphate
<i>-3 ions</i>			
PO_4^{3-}	phosphate	AsO_4^{3-}	arsenate
BO_3^{3-}	borate		

7.3.5 Naming Polyatomic Ions

At first glance, the nomenclature of the polyatomic negative ions in the table above seems hopeless. There are several general rules, however, that can bring some order out of this apparent chaos.

The name of the ion usually ends in either *-ite* or *-ate*. The *-ite* ending indicates a low oxidation state. Thus, the NO_2^- ion is the nitrite ion.

The *-ate* ending indicates a high oxidation state. The NO_3^- ion, for example, is the nitrate ion.

The prefix *hypo-* is used to indicate the very lowest oxidation state. The ClO^- ion, for example, is the hypochlorite ion.

The prefix *per-* (as in hyper-) is used to indicate the very highest oxidation state. The ClO_4^- ion is therefore the perchlorate ion.

There are only a handful of exceptions to these generalizations. The names of the hydroxide (OH^-), cyanide (CN^-), and peroxide (O_2^{2-}) ions, for example, have the *-ide* ending because they were once thought to be monatomic ions.

7.3.6 Naming Simple Covalent Compounds (Non-metals with non-metals)

Oxidation states also play an important role in naming simple covalent compounds. The name of the atom in the positive oxidation state is listed first. The suffix *-ide* is then added to the stem of the name of the atom in the negative oxidation state.

HCl	hydrogen chloride
NO	nitrogen oxide
BrCl	bromine chloride

As a rule, chemists write formulas in which the element in the positive oxidation state is written first, followed by the element(s) with negative oxidation numbers.

The number of atoms of an element in simple covalent compounds is indicated by adding one of the following Greek prefixes to the name of the element.

mono-	
2 di-	7 hepta-
3 tri-	8 octa-
4 tetra-	9 nona-
5 penta-	10 deca-

The prefix *mono-* is seldom used because it is redundant. The principal exception to this rule is carbon monoxide (CO).

7.3.7 Naming Acids

Simple covalent compounds that contain hydrogen, such as HCl, HBr, and HCN, often dissolve in water to produce acids. These solutions are named by adding the prefix *hydro-* to the name of the compound and then replacing the suffix *-ide* with *-ic*. For example, hydrogen chloride (HCl) dissolves in water to form hydrochloric acid; hydrogen bromide (HBr) forms hydrobromic acid; and hydrogen cyanide (HCN) forms hydrocyanic acid.

Many of the oxygen-rich polyatomic negative ions in Table 2.1 form acids that are named by replacing the suffix *-ate* with *-ic* and the suffix *-ite* with *-ous*.

Acids containing ions ending with <i>ide</i> often become			<i>hydro -ic acid</i>
Cl ⁻	chloride	HCl	hydrochloric acid
F ⁻	fluoride	HF	hydrofluoric acid
S ²⁻	sulfide	H ₂ S	hydrosulfuric acid

Acids containing ions ending with <i>ate</i> usually become			<i>-ic acid</i>
CH ₃ CO ₂ ⁻	acetate	CH ₃ CO ₂ H	acetic acid
CO ₃ ²⁻	carbonate	H ₂ CO ₃	carbonic acid
BO ₃ ³⁻	borate	H ₃ BO ₃	boric acid
NO ₃ ⁻	nitrate	HNO ₃	nitric acid
SO ₄ ²⁻	sulfate	H ₂ SO ₄	sulfuric acid
ClO ₄ ⁻	perchlorate	HClO ₄	perchloric acid
PO ₄ ³⁻	phosphate	H ₃ PO ₄	phosphoric acid
MnO ₄ ⁻	permanganate	HMnO ₄	permanganic acid
CrO ₄ ²⁻	chromate	H ₂ CrO ₄	chromic acid
ClO ₃ ⁻	chlorate	HClO ₃	chloric acid

Acids containing ions ending with <i>ite</i> usually become			<i>-ous acid</i>
ClO ₂ ⁻	chlorite	HClO ₂	chlorous acid
NO ₂ ⁻	nitrite	HNO ₂	nitrous acid
SO ₃ ²⁻	sulfite	H ₂ SO ₃	sulfurous acid
ClO ⁻	hypochlorite	HClO	hypochlorous acid

Complex acids can be named by indicating the presence of acidic hydrogen as follows.

NaHCO_3	sodium hydrogen carbonate (also known as sodium bicarbonate)
NaHSO_3	sodium hydrogen sulfite (also known as sodium bisulfite)
KH_2PO_4	potassium dihydrogen phosphate

7.4 Valency

The valency of an atom is the number of single chemical bonds that it can make (in the case of a covalently bonding substance) or the number of electrical charges that it carries (for an ion). Notice that once again the nature of the substance in question requires that the definitions be adapted appropriately. The concept of valence can be used to find the formula of a compound from the valencies of its constituent elements, or to find the valency of an elements within a compound of known formula.

Every atom within a substance is assigned a valency number that is either positive or negative. The total sum of all of the valencies within a formula unit is zero

7.4.1 Using valencies

Once the valencies of a few elements are known it becomes a simple matter to construct the formula of unknown compounds using the valency method. Remember that the sum of the valencies of all of the atoms in the compound must equal zero.

Where an atom may have either positive or negative valency, it is negative if it is the more electronegative element in the compound and positive if not.

Example: From the water molecule above we know that the valency of hydrogen is +1.
 If the valency of nitrogen in ammonia is -3 then we can construct the formula of ammonia thus:
 We need enough hydrogens to cancel out the -3 valency of nitrogen. Each hydrogen = +1 therefore we need three hydrogen atoms.
 The formula of ammonia = NH_3

7.4.2 Working with ions

When using valencies to work out the formula of an ion we have to remember the final charge on the ion must equal the sum of the valencies, taking into account whether the valency of each atom is negative or positive.

Example: Find the formula of the sulfate (2^-) ion given that the valency of the sulfur atom is +VI and the valency of the oxygen atom is -II
Oxygen always has negative valencies (unless bonded to fluorine)
There is one sulfur atom with a valency of +6 and overall the ion has a valency of -2
Therefore $+6 + (xO) = -2$
Therefore $(xO) = -2 - 6 = -8$
each O = -2 therefore there are four oxygen atoms in the ion
Formula of the sulfate ion = **SO_4^{2-}**

EVALUATION

1. Write the symbols and the valencies of the following:
 - i. Iron
 - ii. potassium
 - iii. Oxygen
 - iv. Chlorine
2. What is valency?

STUDY SESSION EIGHT: OXIDATION NUMBER

8.0 INTRODUCTION

An **oxidation number** (also called **oxidation state**) is the number assigned to an atom in a compound or ion to show the apparent loss or gain of electrons by that atom.

Oxidation numbers help us understand how electrons are distributed among the atoms in a compound. They are also useful in identifying **oxidation and reduction reactions**.

For example, in H_2O , hydrogen has an oxidation number of **+1**, while oxygen has an oxidation number of **-2**.

This study session covers:

- The meaning of oxidation numbers.
- The rules for assigning oxidation numbers.
- How to calculate oxidation numbers in compounds and ions.
- The relationship between oxidation numbers and electrons.
- The difference between oxidation and reduction.
- Examples of oxidation numbers in common compounds.

Learning Outcomes:

By the end of this study session, you should be able to:

1. Define an oxidation number.
2. Explain the importance of oxidation numbers in chemical compounds.
3. State the rules for assigning oxidation numbers.
4. Determine the oxidation number of an element in a compound.
5. Calculate the oxidation number of an element in a polyatomic ion.
6. Explain the relationship between oxidation numbers and oxidation-reduction reactions.
7. Identify oxidation and reduction using changes in oxidation numbers.
8. State that:
 - An element in its free state has an oxidation number of **zero (0)**.
 - The oxidation number of a simple ion is equal to its charge.
 - Oxygen is usually **-2**.
 - Hydrogen is usually **+1**.
 - The sum of oxidation numbers in a neutral compound is **zero (0)**.
 - The sum of oxidation numbers in an ion is equal to the **charge of the ion**.
9. Give examples of oxidation numbers in compounds such as H_2O , CO_2 , NaCl , NH_3 , and H_2SO_4 .

8.1 OXIDATION NUMBER

It is often useful to follow chemical reactions by looking at changes in the oxidation numbers of the atoms in each compound during the reaction. Oxidation numbers also play an important role in the systematic nomenclature of chemical compounds. By definition, the **oxidation number** of an atom is the charge that atom would have if the compound was composed of ions.

1. The oxidation number of an atom is zero in a neutral substance that contains atoms of only one element. Thus, the atoms in O_2 , O_3 , P_4 , S_8 , and aluminum metal all have an oxidation number of 0.
2. The oxidation number of simple ions is equal to the charge on the ion. The oxidation number of sodium in the Na^+ ion is +1, for example, and the oxidation number of chlorine in the Cl^- ion is -1.
3. The oxidation number of hydrogen is +1 when it is combined with a *nonmetal* as in CH_4 , NH_3 , H_2O , and HCl .
4. The oxidation number of hydrogen is -1 when it is combined with a *metal* as in LiH , NaH , CaH_2 , and $LiAlH_4$.
5. The metals in Group IA form compounds (such as Li_3N and Na_2S) in which the metal atom has an oxidation number of +1.
6. The elements in Group IIA form compounds (such as Mg_3N_2 and $CaCO_3$) in which the metal atom has a +2 oxidation number.
7. Oxygen usually has an oxidation number of -2. Exceptions include molecules and polyatomic ions that contain O-O bonds, such as O_2 , O_3 , H_2O_2 , and the O_2^{2-} ion.
8. The elements in Group VIIA often form compounds (such as AlF_3 , HCl , and $ZnBr_2$) in which the nonmetal has a -1 oxidation number.
9. The sum of the oxidation numbers in a neutral compound is zero.
$$H_2O: 2(+1) + (-2) = 0$$
10. The sum of the oxidation numbers in a polyatomic ion is equal to the charge on the ion. The oxidation number of the sulfur atom in the SO_4^{2-} ion must be +6, for example, because the sum of the oxidation numbers of the atoms in this ion must equal -2.
$$SO_4^{2-}: (+6) + 4(-2) = -2$$
11. Elements toward the bottom left corner of the periodic table are more likely to have positive oxidation numbers than those toward the upper right corner of the table. Sulfur has a positive oxidation number in SO_2 , for example, because it is below oxygen in the periodic table.
$$SO_2: (+4) + 2(-2) = 0$$

EVALUATION

Calculate the oxidation number of the central elements in the following compounds.

1. $\text{K}_2\text{Cr}_2\text{O}_7$
2. KMnO_4
3. H_2SO_4
4. CrO_7^{2-}

STUDY SESSION NINE: MIXTURES AND SEPERATION TECHNIQUES

9.0 INTRODUCTION

A **mixture** is a substance formed when two or more substances are combined physically without undergoing a chemical reaction. The substances in a mixture retain their individual properties and can usually be separated by **physical methods**.

Mixtures can be classified into **homogeneous mixtures** and **heterogeneous mixtures**. A homogeneous mixture has a uniform composition throughout, while a heterogeneous mixture does not have a uniform composition.

For example, **salt dissolved in water** is a homogeneous mixture, while **sand and water** is a heterogeneous mixture.

The substances that make up a mixture can be separated using different physical techniques. The method used depends on the **physical properties** of the substances involved, such as particle size, boiling point, solubility, density, and magnetism.

This study session covers:

- The meaning of a mixture.
- The differences between elements, compounds, and mixtures.
- The types of mixtures.
- The meaning of separation techniques.
- Different methods used to separate mixtures.
- The principles behind different separation techniques.
- Examples of mixtures and the methods used to separate them.

Learning Outcomes:

By the end of this study session, you should be able to:

1. Define a mixture.
2. Explain the difference between a mixture and a pure substance.
3. Distinguish between homogeneous and heterogeneous mixtures.
4. State the differences between elements, compounds, and mixtures.
5. Explain why the components of a mixture can be separated by physical methods.
6. Identify suitable separation techniques for different mixtures.
7. Explain the principles involved in common separation techniques.
8. State that:
 - **Filtration** is used to separate an insoluble solid from a liquid.
 - **Evaporation** is used to separate a soluble solid from a solution.
 - **Distillation** is used to separate a solvent from a solution or liquids with different boiling points.
 - **Fractional distillation** is used to separate miscible liquids with different boiling points.
 - **Separating funnel** is used to separate immiscible liquids.

- **Chromatography** is used to separate substances based on differences in their solubility and movement through a medium.
 - **Sublimation** is used to separate a sublimable solid from a non-sublimable substance.
 - **Magnetic separation** is used to separate magnetic substances from non-magnetic substances.
9. Give examples of mixtures and the appropriate separation techniques, such as:
- Sand and water — Filtration
 - Salt and water — Evaporation or crystallisation
 - Ethanol and water — Fractional distillation
 - Oil and water — Separating funnel
 - Dyes in ink — Chromatography
 - Iron filings and sulfur — Magnetic separation
 - Ammonium chloride and sand — Sublimation

9.1 MIXTURE

A **mixture** is made from different substances that are **not chemically joined**.

For example powdered iron and powdered sulphur mixed together makes a mixture of iron and sulphur. They can be separated from each other without a chemical reaction, in the way that different coloured sweets can be picked out from a mixed packet and put into separate piles.



9.2 Mixture and compounds

Mixtures have different properties from compounds. The table summarizes these differences.

	Mixture	Compound
Composition	Variable composition – you can vary the amount of each substance in a mixture.	Definite composition – you cannot vary the amount of each element in a compound.
Joined or not	The different substances are not chemically joined together.	The different elements are chemically joined together.
Properties	Each substance in the mixture keeps its own properties.	The compound has properties different from the elements it contains.
Separation	Each substance is easily separated from the mixture.	It can only be separated into its elements using chemical reactions.
Examples	Air, sea water, most rocks.	Water, carbon dioxide, magnesium

9.3 An example - iron, sulphur and iron sulphide

Remember that iron and sulphur react together when they are heated to make a compound called iron sulphide. What are the differences between a mixture of iron and sulphur, and iron sulphide? Here are some of them:

- The mixture can contain more or less iron, but iron sulphide always contains equal amounts of iron and sulphur.
- The iron and sulphur atoms are not joined together in the mixture, but they are joined together in iron Sulphide.
- The iron and sulphur still behave like iron and sulphur in the mixture, but iron sulphide has different properties from both iron and sulphur.
- You can separate the iron from the mixture using a magnet but this does not work for iron Sulphide

9.4 SEPARATION OF MIXTURES USING DIFFERENT TECHNIQUE

9.4.1 Our Objective:

To separate the components of a mixture using the following techniques:

- **Separating funnel**
- **Chromatography**
- **Centrifugation**
- **Simple distillation**
- **Fractional distillation**

9.4.2 The Theory

How is a homogeneous mixture different from a heterogeneous mixture?

Most materials in our surroundings are mixtures of two or more components. Mixtures are either **homogeneous** or **heterogeneous**. **Homogeneous mixtures are uniform in composition**, but **heterogeneous mixtures are not uniform in composition**.

Air is a homogeneous mixture and oil in water is a heterogeneous mixture. Homogeneous and

heterogeneous mixtures can be separated into their components by several physical methods. The choice of separation techniques is based on the type of mixture and difference in the chemical properties of the constituents of a mixture.

What are types of separation techniques?

9.4.5 Various types of separation processes are:

- Crystallization
- Filtration
- Decantation
- Sublimation
- Evaporation
- Simple distillation
- Fractional distillation
- Chromatography
- Centrifugation
- Separating funnel
- Magnetic separation
- Precipitation

Let's discuss some of the separation techniques

Using a separating funnel:

A separating funnel is used for the separation of components of a mixture between two immiscible liquid phases. One phase is the aqueous phase and the other phase is an organic solvent. This separation is based on the differences in the densities of the liquids. The liquid having more density forms the lower layer and the liquid having less density forms the upper layer.

Applications:

- To separate a mixture of oil and water.
- To separate a mixture of kerosene oil and water.

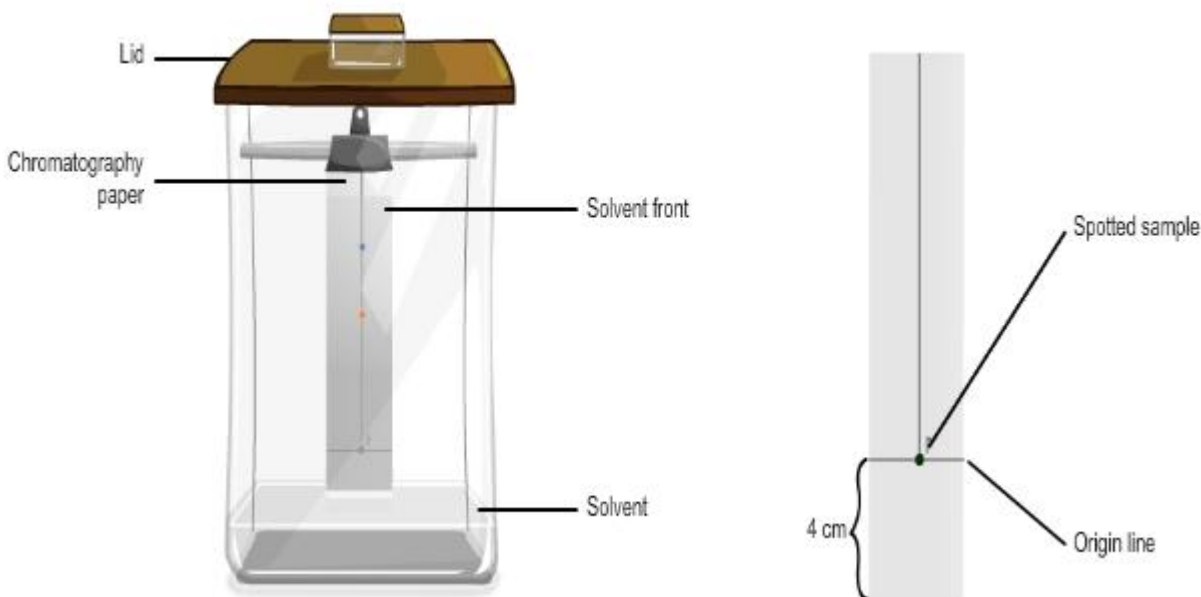
Chromatography:

Chromatography is a separation technique used to separate the different components in a liquid mixture. It was introduced by a Russian Scientist Michael Tswett. Chromatography involves the sample being dissolved in a particular solvent called mobile phase. The mobile phase may be a gas or liquid. The mobile phase is then passed through another phase called stationary phase. The stationary phase may be a solid packed in a glass plate or a piece of chromatography paper.

The various components of the mixture travel at different speeds, causing them to separate. There are different types of chromatographic techniques such as column chromatography, TLC, paper chromatography, and gas chromatography.

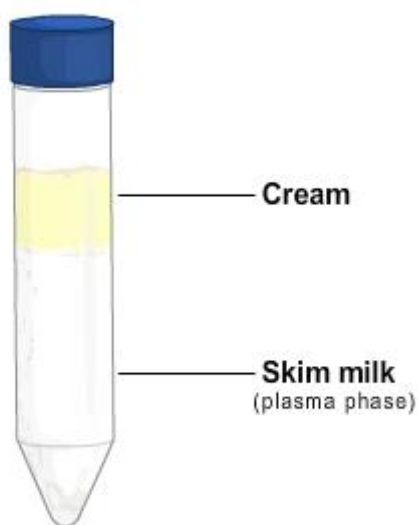
Paper chromatography is one of the important chromatographic methods. Paper chromatography uses paper as the stationary phase and a liquid solvent as the mobile phase. In paper

chromatography, the sample is placed on a spot on the paper and the paper is carefully dipped into a solvent. The solvent rises up the paper due to capillary action and the components of the mixture rise up at different rates and thus are separated from one another.



Applications:

- To separate colors in a dye.
- To separate pigments from natural colors.
- To separate drugs from blood.



Centrifugation:

Sometimes the solid particles in a liquid are very small and can pass through a filter paper. For such particles, the filtration technique cannot be used for separation. Such mixtures are separated by centrifugation. So, **centrifugation is the process of separation of insoluble materials from a liquid where normal filtration does not work well.** The centrifugation is based on the size, shape, and density of the particles, viscosity of the medium, and the speed of rotation. The principle is that the denser particles are forced to the bottom and the lighter particles stay at the top when spun rapidly.

The apparatus used for centrifugation is called a centrifuge. The centrifuge consists of a centrifuge tube holder called rotor. The rotor holds balanced centrifugal tubes of equal amounts of the solid-liquid mixture. On rapid rotation of the rotor, the centrifuge tubes rotate horizontally

and due to the centrifugal force, the denser insoluble particles separate from the liquid. When the rotation stops, the solid particles end up at the bottom of the centrifuge tube with liquid at the top.

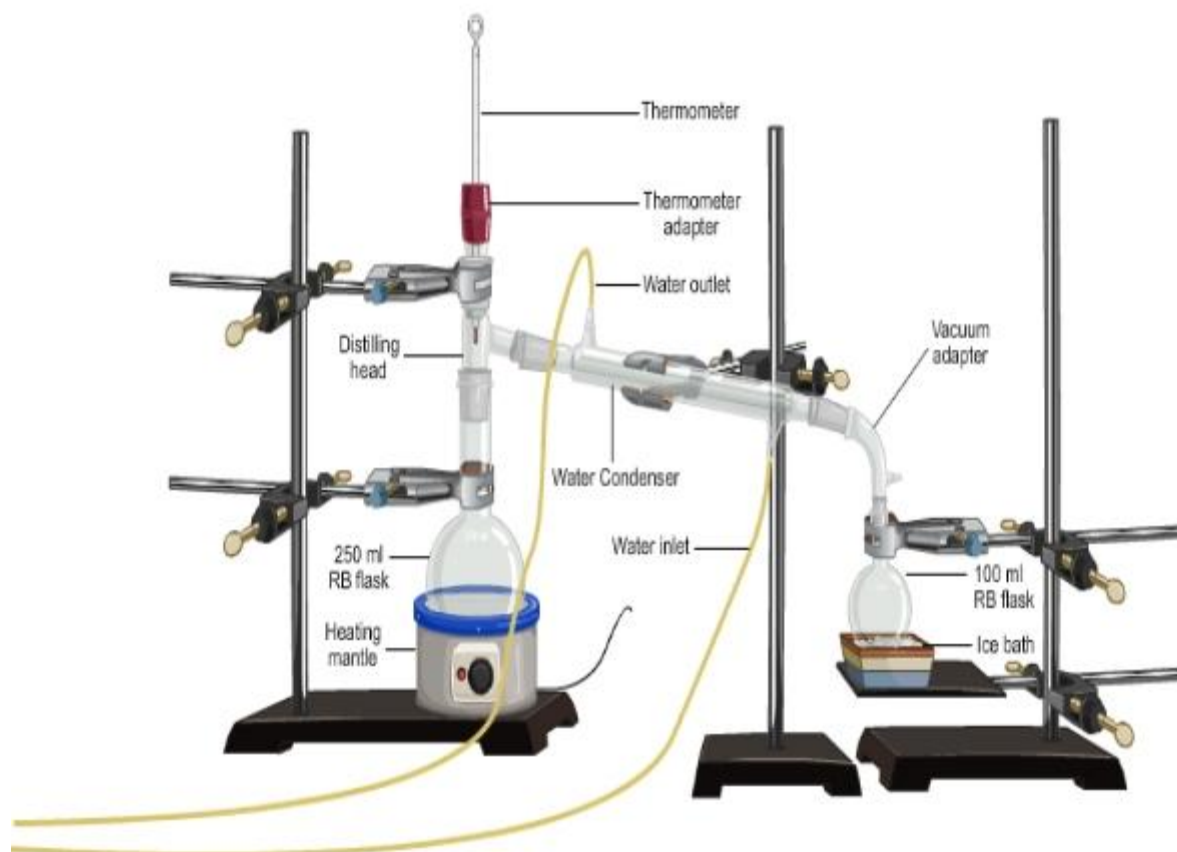
Applications:

- Used in diagnostic laboratories for blood and urine tests.
- Used in dairies and home to separate butter from cream.
- Used in washing machines to squeeze water from wet clothes.

Simple distillation:

Simple distillation is a method used for the separation of components of a mixture containing two miscible liquids that boil without decomposition and have sufficient difference in their boiling points.

The distillation process involves heating a liquid to its boiling points, and transferring the vapors into the cold portion of the apparatus, then condensing the vapors and collecting the condensed liquid in a container. In this process, when the temperature of a liquid rises, the vapor pressure of the liquid increases. When the vapor pressure of the liquid and the atmospheric pressure reach the same level, the liquid passes into its vapor state. The vapors pass over the heated portion of the apparatus until they come into contact with the cold surface of the water-cooled condenser. When the vapor cools, it condenses and passes down the condenser and is collected into a receiver through the vacuum adapter.

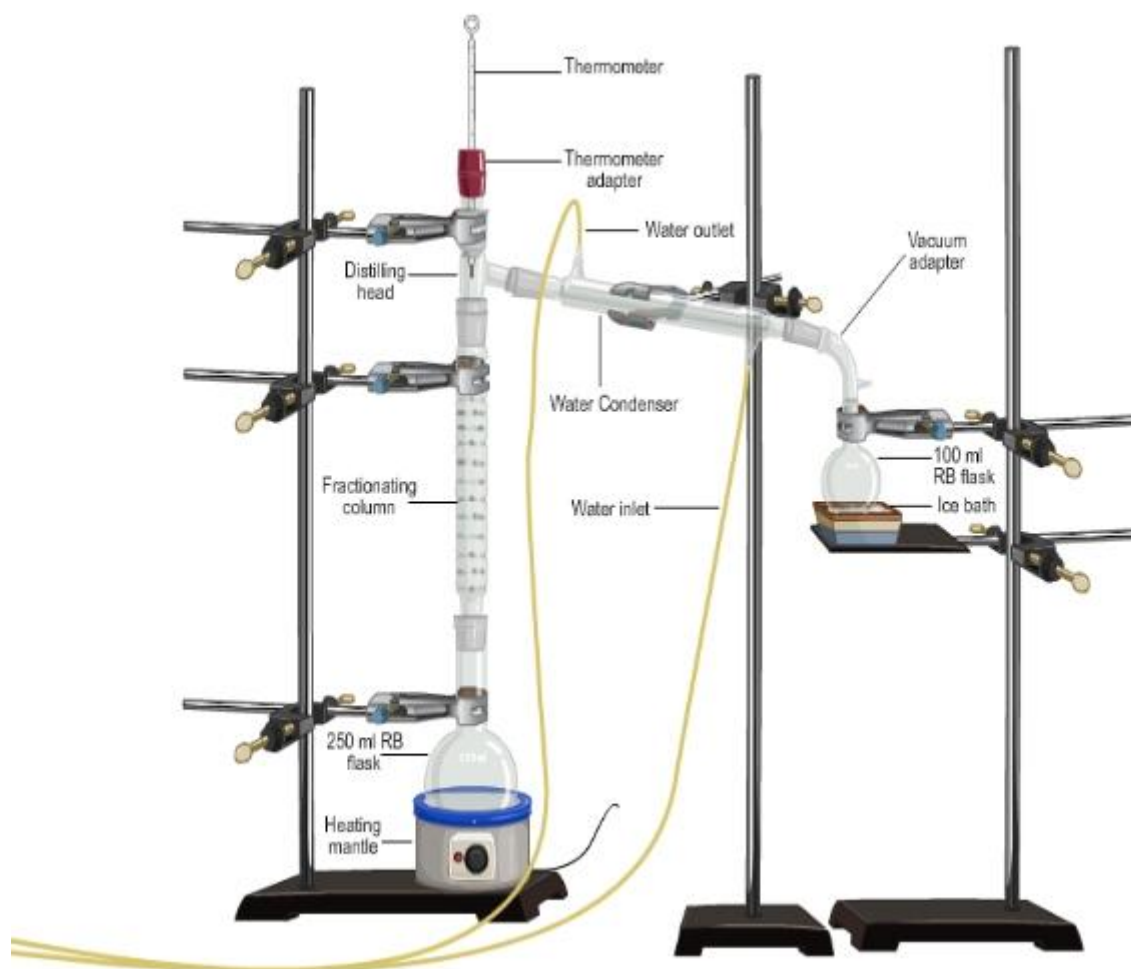


Applications:

- Separation of acetone and water.
- Distillation of alcohol.

Fractional distillation: Fractional distillation is used for the separation of a mixture of two or more miscible liquids for which the difference in boiling points is less than 25K. The apparatus for fractional distillation is similar to that of simple distillation, except that a fractionating column is fitted in between the distillation flask and the condenser.

A simple fractionating column is a tube packed with glass beads. The beads provide surface for the vapors to cool and condense repeatedly. When vapors of a mixture are passed through the fractionating column, because of the repeated condensation and evaporation, the vapors of the liquid with the lower boiling point first pass out of the fractionating column, condense and are collected in the receiver flask. The other liquid, with a slightly higher boiling point, can be collected in similar fashion in another receiver flask.

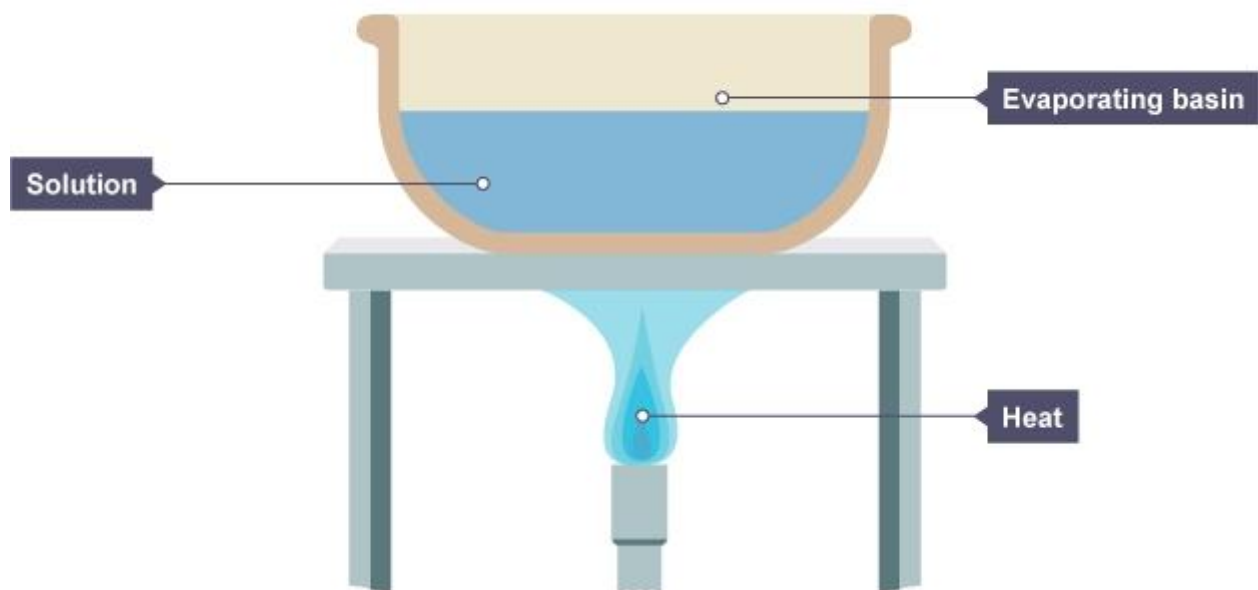


Applications:

- Separation of different fractions from petroleum products.
- Separation of a mixture of methanol and ethanol.

Separating solids from liquids – evaporation

Evaporation is used to separate a soluble solid from a liquid. For example, copper sulfate is soluble in water – its crystals dissolve in water to form copper sulfate solution. During evaporation, the water evaporates away leaving solid copper sulfate crystals behind.



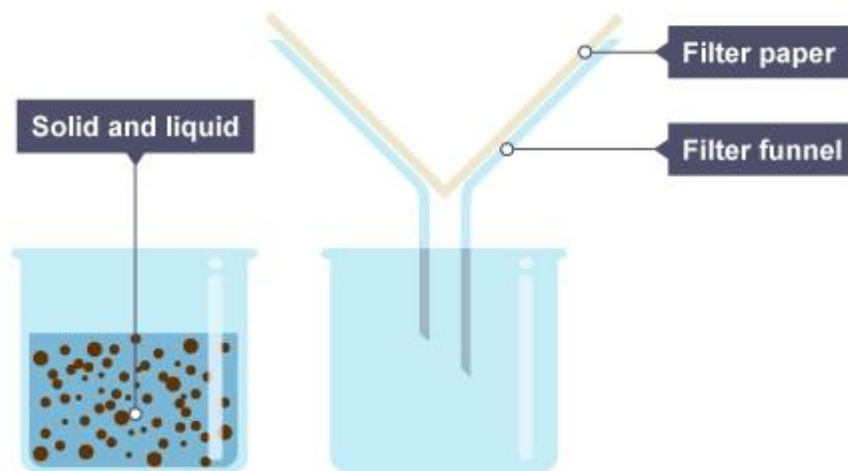
A solution is placed in an evaporating basin and heated with a Bunsen burner.

Separating solids from liquids – filtration

If a substance does not dissolve in a solvent, we say that it is insoluble. For example, sand does not dissolve in water – it is insoluble.

Filtration is a method for separating an insoluble solid from a liquid. When a mixture of sand and water is filtered:

- the sand stays behind in the filter paper (it becomes the residue)
- the water passes through the filter paper (it becomes the filtrate)



A beaker containing a mixture of insoluble solid and liquid. There is filter paper in a filter funnel above another beaker

Separating mixtures

- components in a mixture retain their identities
- exploit properties that distinguish the components to separate mixtures

Some manufacturers add iron filings to cereal to increase its iron content! The bits of iron will stick to a magnet, but the cereal won't. So you can easily separate the mixture by stirring a bar magnet through a slurry of water and finely crushed cereal.

the more similar the properties are, the more difficult it is to separate them

Many elements occur in forms with slightly different masses. For example, uranium occurs as uranium-235, which can be used to construct atomic bombs, and uranium-238, which can't. The two are very difficult to separate because they are nearly identical otherwise. The technical difficulties in separating this mixture are one of the factors that has limited the proliferation of nuclear weapons.

- **Basic Strategies**
 - Phase conversion: convert components of the mixture into other forms that are easy to isolate
 - phase transfer: add a new phase that collects some components from the mixture, but not others

technique	basis for separation	apply this technique to:
adsorption / desorption	phase transfer to a solid surface	liquid or gaseous mixtures that contain at least one component that adsorbs
chromatography	phase transfer from a mobile mixture to a stationary phase	liquid or gaseous solutions that contain several components with differing affinities for the stationary phase
condensation	phase separation by condensing gases in the mixture to liquids	gaseous mixtures containing at least one gas with a much higher boiling point than the others
dialysis	phase transfer through a porous membrane that allows some molecules to pass through, but not others	solutions containing small molecules mixed with very large molecules
effusion	gases with faster molecules flow through tiny pinholes faster than gases with slow molecules	gaseous mixtures containing gases with different molecular weights
dissolution (washing, solvent extraction)	soluble components can be washed away, leaving behind insoluble components (phase transfer to a washing solvent)	mixtures of solids with different solubilities
electrorefining	separate a metal from impurities by dissolving it and then plating it onto an electrode	solid mixtures with a metal as one component
filtration	collect solid particles on a filter	heterogeneous mixture containing a solid phase
floatation	dense components sink, and lighter ones float	heterogeneous mixture with phases with different densities
ion exchange	ions in the mixture bind to surfaces with oppositely charged sites (phase transfer to an ion exchange resin)	solutions containing ions
precipitation	convert solutes to an easily separated solid form	solutions containing a solute that can be precipitated
scrubbing	bubble mixture through a solution that selectively absorbs a component (phase transfer from gas to solution)	gaseous mixtures containing a solute that can be selectively absorbed by a scrubbing solution
stripping	a gas bubbled through the mixture carries off the most volatile components	a liquid mixture containing at least one volatile component

	(phase transfer from solution to gas)	
volatilization (drying, distillation, sublimation)	components with widely differing volatility can be driven out of the mixture by heating (phase change from solid or liquid to gas)	a mixture containing components with differing volatility

Adsorption and desorption

- some solids bind gases and organic materials to their surfaces, removing them from mixtures
- adsorbed gases or liquids can be recovered from the adsorbent material by washing with a solvent
- examples
 - activated charcoal adsorbs many gases and liquids
 - used as a "universal antidote" for poisoning
 - used in water purifiers (removes particulates, lead, copper, mercury, chlorine, hypochlorite, organics)
 - used to adsorb drugs from the blood of overdose victims
 - silica gel absorbs moisture from air

Condensation

- cooling a vapor causes components with the highest boiling points to condense as liquids first
- examples
 - separating steam and air
 - separating oxygen and nitrogen in air

Dialysis

- A semi permeable membrane allows some components in a mixture through, but not others
- How does the membrane distinguish components?
 - Some membranes act as a "molecular sieve" that discriminates between large and small molecules
 - Some membranes dissolve one component better than others
 - Development of new membranes is an active area of research in industry and government
- components flow spontaneously from the high concentration to low concentration side
 - pressure applied to the low concentration side can stop or even reverse this flow (*reverse osmosis*)
- examples
 - purification of blood in dialysis machines
 - purification of seawater by reverse osmosis
 - separation of pollutants from drinking water

Effusion

- use porous membranes to separate light gases from heavy ones
 - average speed of gas molecules depends on the masses of their molecules
 - heavy molecules in a mixture move slower on average than light ones
 - gases made of light molecules diffuse through pores in membranes faster than heavy molecules
- differences from dialysis
 - membrane is permeable, not semi-permeable: all gas molecules in the mixture can pass through it
 - size of molecules isn't usually important: pores in membrane are *much* larger than gas molecules
 - Molecular velocity (and so, molecular mass) is the basis for separation, not size
- examples
 - Separating helium from oxygen
 - Separating uranium isotopes as volatile UF_6

Dissolution (washing)

- separate solids by washing away those that are soluble
- examples
 - separating sand and salt by water washing
 - separating feldspars from quartz in rocks by washing with hot concentrated phosphoric acid
 - separating organic stains from clothing by washing with organic solvents (dry cleaning)

Electro refining

- used to separate metals from impurities
- strategy
 - dissolve the impure metal
 - plate it on an electrode, using a strong electric current
 - pure metal deposits on the electrode, and the impurities stay in solution

Filtration

- pass a mixture that contains solid particles through a porous filter
- if pores are smaller than particles, solid particles stay on filter and liquid/gaseous components pass through
- often used after separation by precipitation

Ion exchange

- used to separate ions from mixtures
- pass the mixture over a surface that is covered with charged sites
- some ions stick to the charged sites
- examples

- water deionization

Precipitation

- precipitation is the conversion of a solute to solid form by chemical or physical change
- solids are then separated by filtration or floatation
- examples
 - separating mud and bacteria from water
 - a gooey aluminum hydroxide precipitate is formed in the water to carry particulates and bacteria to the bottom of a vat
 - clean water is drawn off the top
 - separating sulfate ions from water by adding barium ions
 - barium ion + sulfate = insoluble barium sulfate
 - water softening with washing soda

carbonate + calcium ion = insoluble calcium carbonate

Scrubbing

- scrubbing is bubbling a gas stream through a solution that traps some components
- examples
 - CO₂ can be separated from air by bubbling it through a solution of barium hydroxide
 - H₂S can be removed from air by bubbling it through a zinc acetate solution

Solvent extraction

- a component moves into a solvent shaken with the mixture
 - works best with solvents that dissolve only one component
- Solvent extraction can be used to extract vanillin from vanilla beans. Shaking the beans with an organic solvent like chloroform transfers organic compounds (including the vanillin) to the chloroform. Shaking the chloroform with a sodium hydroxide solution transfers the vanillin into the sodium hydroxide solution.

Stripping

- a stream of gas bubbled through the mixture will carry off the most volatile components
 - Blowing air through a straw in a glass of soda will cause it to go flat, because the air carries off the volatile carbon dioxide.
- volatile components can be reclaimed from the gas by scrubbing

Volatilization

- heating a mixture can cause low-boiling components to volatilize (vaporize)
- several variations
- distillation is collecting and condensing volatilized components

Alcohol can be separated from fermented corn mash by heating the mash to vaporize the alcohol. The vapor is collected and passed through coils of copper tubing, where it cools and condenses. The condensed alcohol is then collected and passed through a series of stills to purify it.

condenses as a liquid once again. Moonshiners sometimes used old car radiators for the condensation step; the soldered joints added a toxic quantity of lead to the shine!

- drying is complete volatilization of some components in the mixture

Separation of water from clothes on a clothesline is one obvious example. The separation of salt from seawater using evaporating pools is another.

- sublimation is volatilization of a solid (without melting!)

Dry ice (solid carbon dioxide) is probably the most familiar example of a solid that sublimates. But water ice can also be converted directly into water vapor without melting, at low pressure. Snow on mountain peaks disappears without moistening the soil.

Separation by sublimation is sometimes called 'freeze drying'. Instant coffee is manufactured by freeze drying. (Boiling the coffee destroys the delicate molecules that give coffee its flavor, and so does exposure to air after a certain time, so distillation or simple drying isn't used). Fresh coffee is frozen to form a mixture of ice and coffee crystals. The pressure over the mixture is lowered so that the ice sublimates, leaving the coffee crystals behind.

EVALUATION

1. Define and give one example each of i. A homogenous mixture and ii. A heterogeneous mixture.
2. Define the following terms. i. filtrate ii. distillate iii. sublimate
3. How would you separate a mixture of salt and sand? Explain.
4. How would you separate the component of black ink?
5. List and explain all the separation technique. Describe their industrial application.